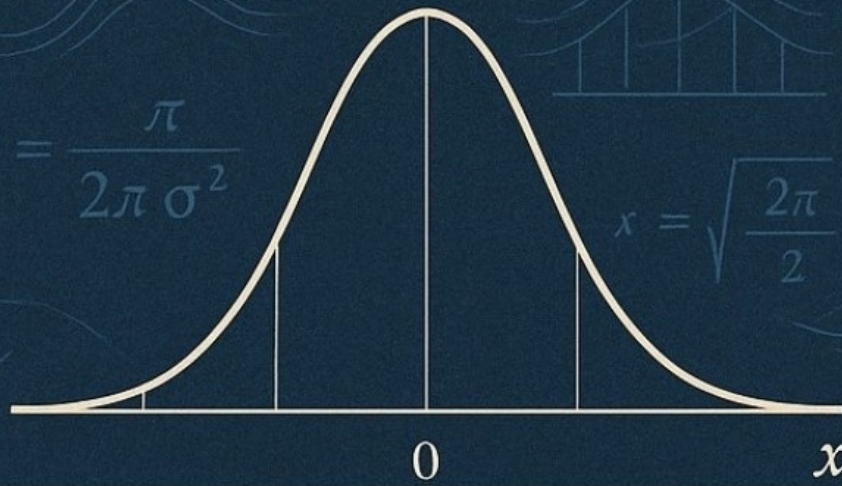


الإجمال في نظرية الاحتمال بطلاء من علم الإحصاء



$P(\text{bad thing happening} \mid \text{you are a Muslim}) = \text{whatever}$

المقدمة

١-أَوَّلُ مَا نَسْتَفْتِحُ الْمَقَالَا

بِذِكْرِ حَمْدِ رَبِّنَا تَعَالَى

٢-فَالْحَمْدُ لِلَّهِ عَلَى مَا أَنْعَمَا

حَمْدًا بِهِ يَجْلُو عَنِ الْقَلْبِ الْعَمَى

٣-ثُمَّ الصَّلَاةُ بَعْدُ وَالسَّلَامُ

عَلَى نَبِيِّ دِينِهِ الْإِسْلَامُ

٤-مُحَمَّدٌ خَاتَمِ رُسُلِ رَبِّهِ

وَأَلِهِ مِنْ بَعْدِهِ وَصَحْبُهُ

٥-وَنَسْأَلُ اللَّهَ لَنَا الْإِعَانَةَ

فِيمَا تَوَخَّيْنَا مِنَ الْإِبَانَةِ

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Common Set Operations and De Morgan's Laws

1. Union

$$A \cup B = \{x \mid x \in A \text{ or } x \in B\}$$

2. Intersection

$$A \cap B = \{x \mid x \in A \text{ and } x \in B\}$$

3. Set Difference

$$A - B = \{x \mid x \in A \text{ and } x \notin B\}$$

4. Complement

$$A^c = \{x \mid x \notin A\}$$

$$A^c = \Omega - A \quad (\text{relative to universal set } \Omega)$$

5. Symmetric Difference

$$A \Delta B = (A - B) \cup (B - A)$$

$$A \Delta B = (A \cup B) - (A \cap B)$$

6. Subset and Superset

$$A \subseteq B \iff \forall x(x \in A \Rightarrow x \in B)$$

$$A \supseteq B \iff \forall x(x \in B \Rightarrow x \in A)$$

7. Equality of Sets

$$A = B \iff A \subseteq B \text{ and } B \subseteq A$$

8. Cartesian Product

$$A \times B = \{(a, b) \mid a \in A, b \in B\}$$

9. Power Set

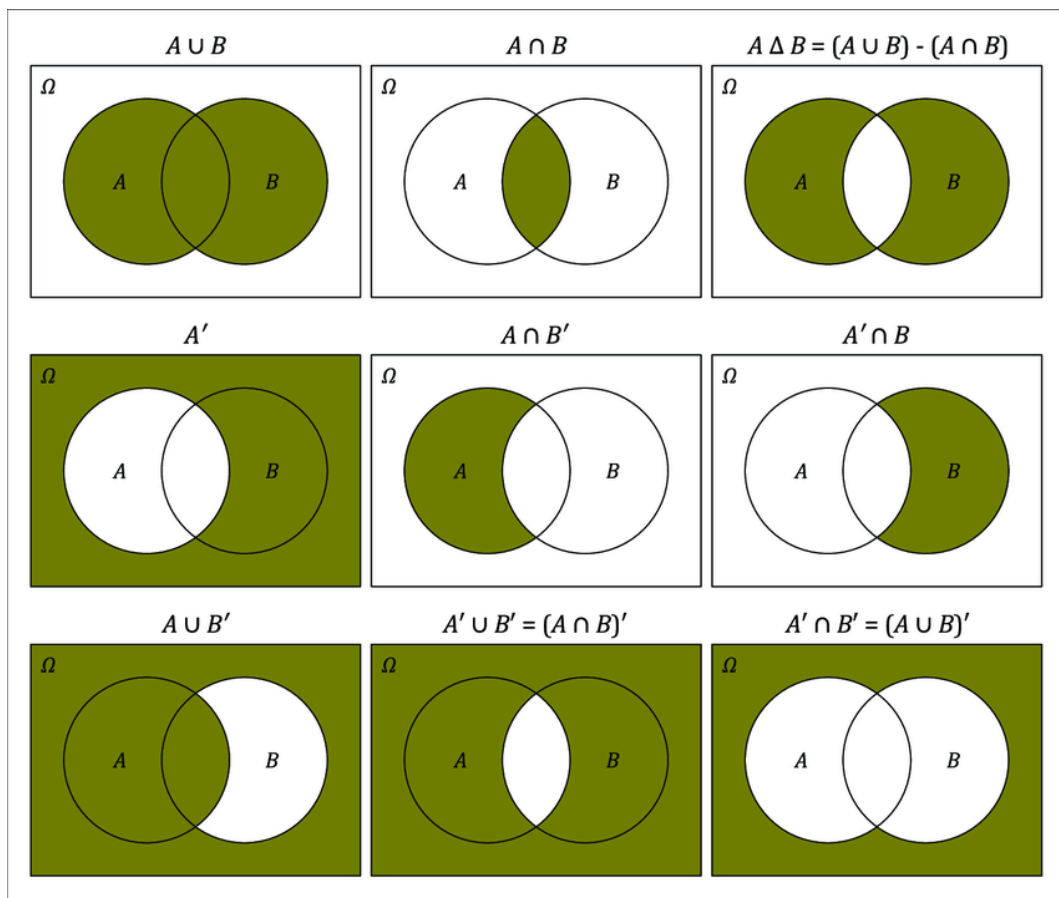
$$\mathcal{P}(A) = \{S \mid S \subseteq A\}$$

De Morgan's Laws

$$(A \cup B)^c = A^c \cap B^c$$

$$(A \cap B)^c = A^c \cup B^c$$

A Venn diagram of unions and intersections for two sets, A and B and their complements



Probability Axioms

1. $P(A) \geq 0$
2. $P(\Omega) = 1$
3. $P(A \cup B) = P(A) + P(B)$ if $A \cap B = \emptyset$

Basic probabilities rules

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A \cap B) = 0 \quad \text{if } A \text{ and } B \text{ are disjoint (mutually exclusive)}$$

$$P(A \cap B) = P(A) \cdot P(B) \quad (\text{if } A \text{ and } B \text{ are independent})$$

$$P(B | A) = P(B) \quad \text{and} \quad P(A | B) = P(A) \quad \text{if } A \text{ and } B \text{ are independent}$$

$$P(A \cap B) = P(A) \cdot P(B|A) = P(B) \cdot P(A|B) \quad (\text{if dependent})$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad (\text{conditional probability})$$

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

$$P(A \cap B | C) = \text{Probability that both } A \text{ and } B \text{ occur, given } C$$

$$P(A \cup B \cup C) = P(A) + P(B) + P(C)$$

$$- P(A \cap B) - P(B \cap C) - P(C \cap A) + P(A \cap B \cap C)$$

◆ 1. Disjoint (Mutually Exclusive) Events

Example:

- Rolling a die:
 - Let $A = \{\text{rolling a 2}\}$
 - Let $B = \{\text{rolling a 5}\}$

You can't roll both a 2 and a 5 at the same time $\rightarrow A \cap B = \emptyset$

✓ These are **disjoint (mutually exclusive)** events.

NOTE: All disjoint events are **dependent**, because knowing one happened tells you the other did not!

◆ 2. Independent Events

Example:

- Toss a coin and roll a die:
 - Let $A = \{\text{coin shows heads}\}$
 - Let $B = \{\text{die shows a 6}\}$

They don't affect each other, so:

✓ These are **independent** events.

NOTE: Independent events can **overlap**. That's a key difference from disjoint events.



Bayes' Theorem - Picnic Decision Example

$$P(A | B) = \frac{P(B | A) \cdot P(A)}{P(B)}$$

You're planning a picnic for your family. You're trying to decide whether to **postpone due to rain**.

- The chance of **rain** on any day is 15%.
- The morning of the picnic, it's **cloudy**.
- The probability of it being **cloudy** is 25%.
- On days where it **rains**, it's cloudy in the morning **80% of the time**.

Should you postpone the picnic?



Given:

- $P(\text{rain}) = 0.15$
- $P(\text{cloudy}) = 0.25$
- $P(\text{cloudy} | \text{rain}) = 0.80$



Bayes' Theorem:

$$P(A | B) = \frac{P(B | A) \cdot P(A)}{P(B)}$$

In our case:

$$P(\text{rain} | \text{cloudy}) = \frac{P(\text{cloudy} | \text{rain}) \cdot P(\text{rain})}{P(\text{cloudy})}$$



Substituting values:

$$\begin{aligned} P(\text{rain} | \text{cloudy}) &= \frac{0.8 \cdot 0.15}{0.25} \\ &= \frac{0.12}{0.25} = 0.48 \end{aligned}$$



Conclusion:

$$P(\text{rain} | \text{cloudy}) = 0.48$$

There is a **48% chance of rain** given that it's cloudy.
You might want to consider **postponing the picnic**!

◆ Probability vs. Likelihood

Probability

- **Conceptually:** Measures the chance of an event occurring, given fixed model parameters.
- It is a function of **outcomes**.
- Always between **0 and 1**.

Mathematically:

$$P(E) = \frac{\text{Number of favorable outcomes}}{\text{Total outcomes}}$$

or more generally,

$$P(x \mid \theta)$$

where θ are known parameters and x is the random outcome.

Likelihood

- **Conceptually:** Measures how plausible different parameter values are, given observed data.
- It is a function of **parameters**, not outcomes.
- Used mainly in estimation (e.g., Maximum Likelihood Estimation).

Mathematically:

$$L(\theta \mid x) = P(x \mid \theta)$$

- Same formula as probability, but the perspective is reversed:
 - **Probability:** parameters fixed, outcome varies.
 - **Likelihood:** outcome fixed (observed), parameters vary.
-

◆ Probability vs. Odds

Probability

- **Conceptually:** The chance of an event happening.
- Always between **0 and 1**.
- Example: Probability of rolling a 6 on a fair die = $\frac{1}{6} \approx 0.167$.

Mathematically:

$$P(E) = \frac{\text{Number of favorable outcomes}}{\text{Total outcomes}}$$

Odds

- **Conceptually:** The ratio of the probability of success to the probability of failure.
- Ranges from **0 to ∞** (not limited to 1).
- Example: Odds of rolling a 6 = $\frac{1}{6} / \frac{5}{6} = \frac{1}{5} = 0.2$.

Mathematically:

$$\text{Odds}(E) = \frac{P(E)}{1 - P(E)}$$

◆ Conversion Between Probability and Odds

- From **probability to odds**:

$$\text{Odds} = \frac{P}{1 - P}$$

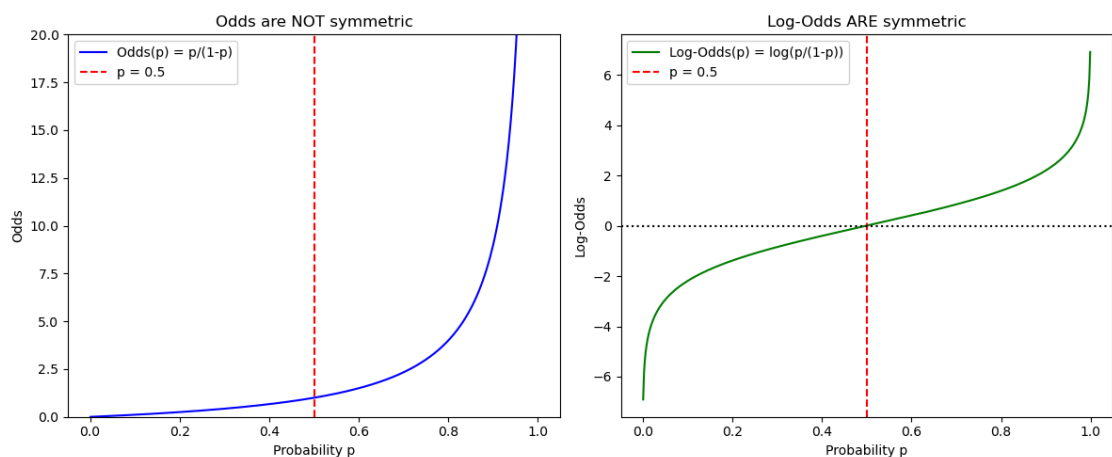
- From **odds to probability**:

$$P = \frac{\text{Odds}}{1 + \text{Odds}}$$

◆ Why Log-Odds?

- The **log of odds** is called the **logit**:

$$\text{logit}(P) = \ln\left(\frac{P}{1 - P}\right)$$



Reasons:

1. Symmetry:

- Odds range $0 \rightarrow \infty$.
- Log-odds range $-\infty \rightarrow +\infty$, which is much easier for modeling with linear regression.

2. Linear Relationship:

- Logistic regression models the **log-odds as a linear function** of predictors:

$$\ln\left(\frac{P}{1-P}\right) = \beta_0 + \beta_1 x_1 + \dots + \beta_k x_k$$

3. Probability Recovery:

- Applying the inverse transformation (sigmoid) gives the probability:

$$P = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \dots + \beta_k x_k)}}$$

◆ Logistic Regression Connection

- Logistic regression doesn't predict probabilities directly.
- It predicts **log-odds**, which can take any real value.
- Then, it maps log-odds back to **probabilities** using the **sigmoid function**:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

✓ In short:

- **Probability** = "chance of success".
- **Odds** = "success-to-failure ratio".
- **Log-odds** = "linear, unbounded version of odds" → makes logistic regression possible.

Counting Principles in Combinatorics

1. Factorial

Definition:

The factorial of a number (n), written ($n!$), is the number of ways to arrange (n) distinct items in order.

Formula:

$$n! = n \cdot (n - 1) \cdot (n - 2) \cdots 2 \cdot 1$$

Example:

If you have 4 different books and want to know how many ways to arrange them on a shelf:

$$4! = 4 \cdot 3 \cdot 2 \cdot 1 = 24 \text{ ways}$$

2. Permutations (Ordered Selection)

Definition:

Permutations count the number of **ordered** arrangements of (r) items chosen from a set of (n) distinct items.

- **Without Replacement** (no repetition allowed):

$$P(n, r) = \frac{n!}{(n - r)!}$$

Example:

Choosing 3 winners (gold, silver, bronze) from 5 runners:

$$P(5, 3) = \frac{5!}{(5 - 3)!} = \frac{120}{2} = 60 \text{ ways}$$

- **With Replacement** (repetition allowed):

$$n^r$$

Example:

Choosing a 4-digit PIN code from digits 0–9 (with digits allowed to repeat):

$$10^4 = 10,000 \text{ possible PINs}$$

3. Combinations (Unordered Selection)

Definition:

Combinations count the number of **unordered** selections of (r) items from a set of (n) distinct items.

- **Without Replacement** (no repetition allowed):

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}$$

Example:

Choosing 3 students from a class of 10 for a project group:

$$\binom{10}{3} = \frac{10!}{3! \cdot 7!} = 120 \text{ ways}$$

- **With Replacement** (repetition allowed):

$$\binom{n+r-1}{r}$$

Example:

Choosing 3 scoops of ice cream (flavors can repeat) from 5 flavors:

$$\binom{5+3-1}{3} = \binom{7}{3} = 35 \text{ combinations}$$



Subsets and Powers of Two

Let's look at the **number of subsets** of an (n)-element set.

The total number of **subsets** is the sum over all (r)-combinations:

$$\sum_{r=0}^n \binom{n}{r} = 2^n$$

This makes sense because for each of the (n) elements, we can choose to either **include** or **exclude** it from a subset.

Proof using extreme cases:

- When $r = 0$, $\binom{n}{0} = 1 \rightarrow$ only the empty set
- When $r = n$, $\binom{n}{n} = 1 \rightarrow$ only the full set

Hence, total number of subsets is:

$$\binom{n}{0} + \binom{n}{1} + \cdots + \binom{n}{n} = 2^n$$



4. Multiplication (Product) Rule

Definition:

If a process consists of k independent steps, and the i -th step can be performed in n_i ways, then the total number of possible outcomes is:

$$\prod_{i=1}^k n_i$$

$$n_1 \cdot n_2 \cdot \cdots \cdot n_k$$

Example:

Creating an outfit:

- 3 shirts
- 2 pants
- 2 shoes

Total combinations:

$$3 \cdot 2 \cdot 2 = 12 \text{ outfits}$$

 5. Partitions — Multinomial Coefficients

Definition:

The number of ways to divide n distinct items into k labeled groups of sizes $n_1, n_2, \dots, n_k$ (where $n_1 + n_2 + \dots + n_k = n$) is:

$$\frac{n!}{n_1! \cdot n_2! \cdot \dots \cdot n_k!}$$
$$\binom{n}{n_1, n_2, \dots, n_k} = \frac{n!}{n_1! \cdot n_2! \cdot \dots \cdot n_k!}$$

This is called the **multinomial coefficient** and it generalizes the binomial coefficient.

Example:

How many ways to split 7 people into 3 groups:

- 2 in team A
- 3 in team B
- 2 in team C:

$$\binom{7}{2, 3, 2} = \frac{7!}{2! \cdot 3! \cdot 2!} = \frac{5040}{2 \cdot 6 \cdot 2} = 210$$

Special Case:

When ($k = 2$), the multinomial becomes a binomial:

$$\binom{n}{r, n-r} = \frac{n!}{r! \cdot (n-r)!} = \binom{n}{r}$$

Concept	Use Permutations $P(n, r)$	Use Combinations $\binom{n}{r}$
Order matters	✔ Yes	✘ No
Order doesn't matter	✘ No	✔ Yes
Items are distinct	✔ Yes	✔ Yes
No repetition (default)	✔ Usually	✔ Usually

 Standard 52-Card Deck Overview

Structure:

- **Total cards:** 52
 - **Suits (4):**
 - ♥ Hearts (red)
 - ♦ Diamonds (red)
 - ♣ Clubs (black)
 - ♠ Spades (black)
 - **Ranks (13 per suit):** A, 2, 3, ..., 10, J, Q, K
-

Summary:

-  Number cards (2–10): $9 \times 4 = 36$
 -  Face cards (J, Q, K): $3 \times 4 = 12$
 -  Aces: 1 per suit = 4
 -  Red cards: 26 (♥ + ♦)
 -  Black cards: 26 (♣ + ♠)
-

Probability: Each Player Gets One Ace

Problem:

A 52-card deck is dealt **evenly** to 4 players (13 cards each).
What is the probability that **each player gets exactly one Ace**?

Solution (Partitioning Approach):

- Total number of fair deals (unordered hands of 13 cards):

$$\text{Total} = \frac{52!}{(13!)^4}$$

- Favorable outcomes:
 - Assign 1 Ace to each hand: $4!$
 - Partition the remaining 48 cards into 4 hands of 12 cards:

$$\frac{48!}{(12!)^4}$$

- So the probability is:

$$P = \frac{4! \cdot \frac{48!}{(12!)^4}}{\frac{52!}{(13!)^4}} = \frac{4! \cdot 48! \cdot (13!)^4}{52! \cdot (12!)^4}$$

Multinomial Probability Formula

Concept:

Used when an experiment has:

- A fixed number of trials n
- Each trial results in one of r possible outcomes
- Probabilities for each outcome: $p_1, p_2, \dots, p_r$, where:

$$\sum_{i=1}^r p_i = 1$$

You want to compute the probability of getting:

- n_1 times outcome 1
- n_2 times outcome 2
- ...
- n_r times outcome r ,

where $n_1 + n_2 + \dots + n_r = n$

Multinomial Probability Formula:

$$P(n_1, n_2, \dots, n_r) = \frac{n!}{n_1! n_2! \dots n_r!} \cdot p_1^{n_1} p_2^{n_2} \dots p_r^{n_r}$$

Example: Colored Balls

You draw 5 balls **with replacement**. Each ball is:

-  Red with probability $p_1 = 0.2$
-  Green with probability $p_2 = 0.3$
-  Blue with probability $p_3 = 0.5$

What is the probability of getting:

- 1 Red
- 2 Green
- 2 Blue

Apply the Formula:

Given: $n = 5, n_1 = 1, n_2 = 2, n_3 = 2$

$$\begin{aligned} P(1, 2, 2) &= \frac{5!}{1! \cdot 2! \cdot 2!} \cdot 0.2^1 \cdot 0.3^2 \cdot 0.5^2 \\ &= \frac{120}{1 \cdot 2 \cdot 2} \cdot (0.2)(0.09)(0.25) \\ &= 30 \cdot 0.0045 = \boxed{0.135} \end{aligned}$$

Descriptive Statistics

Types of Data (Levels of Measurement)

In statistics and data science, data are categorized into different types (also called *levels of measurement*).

These determine what kind of analysis and operations are valid.

1. Qualitative Data (Categorical Data)

Nominal Data

- Pure labels/categories, no inherent order.
 - **Examples:** gender (male/female), city names, colors, fruit types.
 - **Other common names:**
 - Labels
 - Discrete categories
 - Unordered categories
 - Classification data
-

Ordinal Data

- Categories **with order**, but the intervals between them are not consistent or meaningful.
 - **Examples:** satisfaction rating (*poor* → *fair* → *good* → *excellent*), education level, ranks in a race.
 - **Other common names:**
 - Ranked data
 - Ordered categories
 - Hierarchical categorical data
-

2. Quantitative Data (Numeric Data)

Interval Data

- Numeric values with meaningful differences, but **no true zero**.
 - **Examples:** temperature in °C/°F, calendar years (1990, 2000, 2010).
 - **Other common names:**
 - Scaled data
 - Continuous (without true zero)
 - Additive numeric data
-

Ratio Data

- Numeric values with meaningful differences **and a true zero** (so ratios make sense).
 - **Examples:** height, weight, age, distance, income.
 - **Other common names:**
 - Measurement data
 - Continuous data with absolute zero
 - True-scale data
-

3. Other Useful Classifications

- **Discrete Data**
 - Countable, finite values.
 - **Examples:** number of students in a class, number of cars, shoe size.
 - **Synonyms:** count data, integer data, finite numeric
 - **Continuous Data**
 - Can take *any value within a range*.
 - **Examples:** height, weight, speed, Kelvin temperature.
 - **Synonyms:** measurement data, real-valued data, interval/ratio data
-



Summary

- **Nominal:** the data can only be categorized
 - *Examples:* blood type (A, B, AB, O), country names
 - **Ordinal:** the data can be categorized and ranked
 - *Examples:* class rank (1st, 2nd, 3rd), satisfaction level (low, medium, high)
 - **Interval:** the data can be categorized, ranked, and evenly spaced
 - *Examples:* temperature in Celsius/Fahrenheit, IQ scores
 - **Ratio:** the data can be categorized, ranked, evenly spaced, and has a natural zero
 - *Examples:* age, weight, income
-



Mean, Trimmed Mean, Weighted Mean, Median, and Mode



1. Mean (Arithmetic Average)

Definition:

$$\text{Mean} = \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

Example:

$$x = [3, 5, 8, 10] \Rightarrow \bar{x} = \frac{3 + 5 + 8 + 10}{4} = 6.5$$

When to use:

- Symmetric data
- No extreme outliers
- Parametric inference (e.g., Central Limit Theorem)

Disadvantages:

- Highly sensitive to outliers
-

2. Trimmed Mean

1) What is the Trimmed Mean?

The **trimmed mean** is an average computed after removing a certain percentage of the smallest and largest values from a dataset.

It is less sensitive to **outliers** than the regular mean.

2) Why is it Useful?

- Reduces the effect of **extreme values (outliers)**.
 - Provides a better **measure of central tendency** for skewed distributions.
 - Commonly used in **sports scoring, economics, and robust statistics**.
-

3) Formula

Let the sorted data be:

$$x_{(1)}, x_{(2)}, \dots, x_{(n)}.$$

If we trim a fraction α from each side, then:

$$k = \alpha \cdot n$$

The **trimmed mean** is:

$$\bar{x}_{\text{trim}} = \frac{1}{n - 2k} \sum_{i=k+1}^{n-k} x_{(i)}.$$

4) Percentage Trimmed

- Common choices: 5%, 10%, 20%.

- Example: **10% trimmed mean** → remove lowest 10% and highest 10%.
-

5) Real-World Example

Exam scores for 10 students:

$$\{40, 45, 50, 55, 60, 65, 70, 75, 95, 100\}$$

- $n = 10$ values
- Trim 10% $\Rightarrow k = 0.1 \cdot 10 = 1$
- Remove the lowest (40) and highest (100)

Remaining values:

$$\{45, 50, 55, 60, 65, 70, 75, 95\}$$

Trimmed mean:

$$\bar{x}_{\text{trim}} = \frac{45 + 50 + 55 + 60 + 65 + 70 + 75 + 95}{8} = \frac{515}{8} = 64.375$$

✓ Interpretation:

- Regular mean of all 10 scores = 65.5
 - 10% trimmed mean = 64.375
 - The trimmed mean is less influenced by the outliers (40 and 100).
-

3. Weighted Mean

Definition:

$$\bar{x}_w = \frac{\sum_{i=1}^n w_i x_i}{\sum_{i=1}^n w_i}$$

Example:

$$x = [3, 5, 8], \quad w = [2, 1, 2] \Rightarrow \bar{x}_w = \frac{2 \cdot 3 + 1 \cdot 5 + 2 \cdot 8}{2 + 1 + 2} = \frac{6 + 5 + 16}{5} = 5.4$$

When to use:

- When observations have different importance (e.g., GPA, financial indices)

Disadvantages:

- Sensitive to extreme weights or values
-

4. Median

Definition: The middle value after sorting the data.

How to compute:

- If n is odd: middle value
- If n is even: average of the two middle values

Examples:

- $x = [1, 4, 6] \Rightarrow \text{Median} = 4$
- $x = [1, 4, 6, 8] \Rightarrow \text{Median} = \frac{4+6}{2} = 5$

When to use:

- Skewed distributions
- Ordinal data
- Robust against outliers

Disadvantages:

- Ignores the exact value of most data points

5. Mode

Definition: The value(s) that appear most frequently.

Examples:

- $x = [1, 2, 2, 3, 4] \Rightarrow \text{Mode} = 2$
- $x = [1, 2, 2, 3, 3] \Rightarrow \text{Modes} = 2, 3$

When to use:

- Categorical data
- Find the most common label

Disadvantages:

- May not be unique
- May not exist in continuous data

Summary Table

Measure	Best For	Disadvantage
Mean	Symmetric distributions, parametric stats	Sensitive to outliers
Trimmed Mean	Reducing the effect of outliers by trimming tails	Data loss; choice of trimming percentage is arbitrary
Weighted Mean	Averages where values have different importance	Still sensitive to extreme weights or values
Median	Skewed distributions, ordinal data	Ignores most values, less efficient for small n

Measure	Best For	Disadvantage
Mode	Categorical data, most frequent value	Might not exist or might be multiple

6. Quantiles and Percentiles

Concept of Quantiles and Percentiles

- A **percentile** is a value below which a given percentage of data falls.
- A **quantile** divides the data into equal-sized intervals.

If we denote:

- $P_k = k^{\text{th}}$ percentile $\rightarrow k\%$ of data lies **below** this value.
- $Q_p = p^{\text{th}}$ quantile $\rightarrow p$ is between 0 and 1

Then:

- $Q_1 = 25^{\text{th}}$ percentile = first quartile
- $Q_2 = 50^{\text{th}}$ percentile = second quartile = **median**
- $Q_3 = 75^{\text{th}}$ percentile = third quartile

Quartiles as a Special Case of Quantiles

When dividing the dataset into **4 equal parts**, we call the quantiles **quartiles**:

- Q_1 : 25% of data lies below this point
- Q_2 : 50% of data lies below this point (the median)
- Q_3 : 75% of data lies below this point

Quartile Example (Even number of data points)

Given sorted data:

$$x = [2, 4, 5, 8, 9, 12, 15, 18]$$

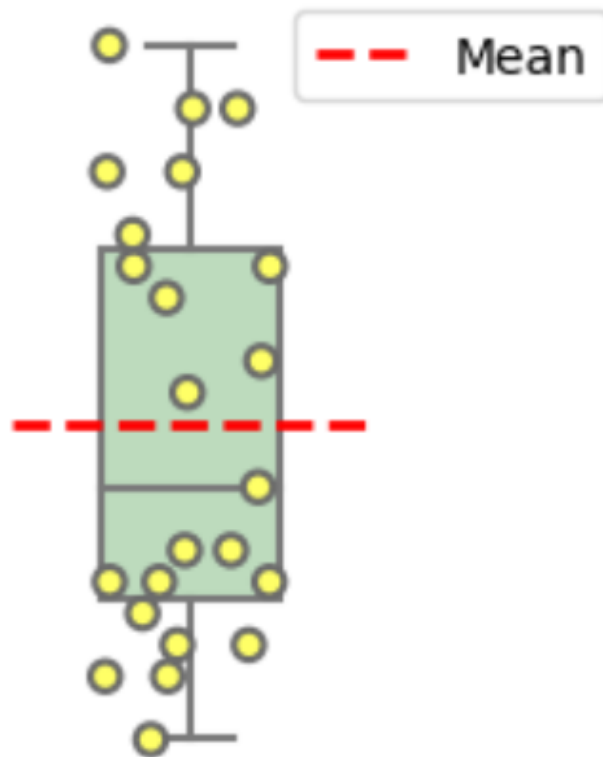
- Lower half: $[2, 4, 5, 8] \Rightarrow Q_1 = \frac{4+5}{2} = 4.5$
- Upper half: $[9, 12, 15, 18] \Rightarrow Q_3 = \frac{12+15}{2} = 13.5$
- Median (Q_2): $Q_2 = \frac{8+9}{2} = 8.5$

Summary

Quantiles divide data into equal-sized intervals. When we divide the data into 100 equal parts, the quantiles are called percentiles. For example, the 0.40 quantile is the same as the 40th percentile, or 40% of the data falls below this value.

When we divide the data into 4 equal parts, the quantiles are called quartiles, such as Q_1 (25th percentile), Q_2 (50th percentile), and Q_3 (75th percentile).

Boxplot and the IQR (Interquartile Range)



Boxplot and the IQR (Interquartile Range)

The **boxplot** is a visual representation of the 5-number summary:

- The lower end of the box
- Q_1
- Median (Q_2)
- Q_3
- The upper end of the box

Key concepts:

- Box height = IQR = $Q_3 - Q_1 \rightarrow$ contains the middle 50% of the data
 - Whiskers extend to:
 - Lower bound: $Q_1 - 1.5 \times \text{IQR}$
 - Upper bound: $Q_3 + 1.5 \times \text{IQR}$
 - Outliers beyond whiskers are shown as dots
 - Mean can optionally be shown with a dashed line or symbol
 - Median is shown as a solid line inside the box
-



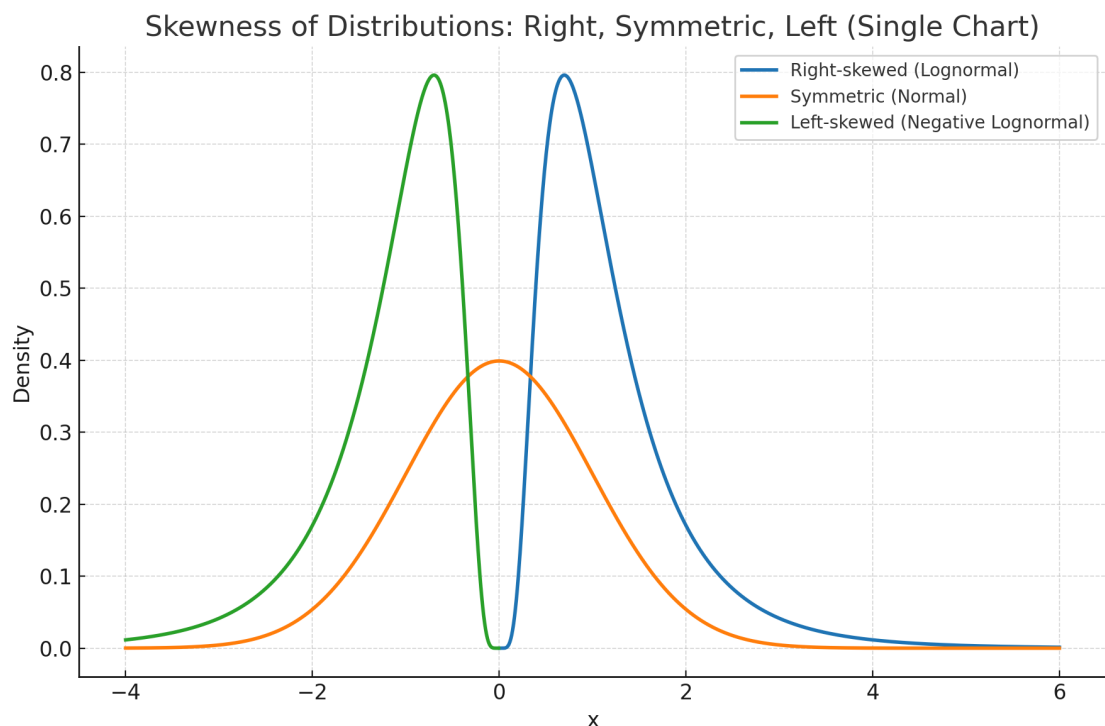
Skewness of a Distribution



What is Skewness?

Skewness measures the **asymmetry** of a distribution:

- Right tail longer $\Rightarrow$ **Right-skewed** (positive skew)
- Left tail longer $\Rightarrow$ **Left-skewed** (negative skew)
- Symmetric $\Rightarrow$ **No skew**



Types of Skewness

► Positive Skew (Right-Skewed)

- Tail on the **right**
- Most values are small
- **Skewness** > 0

$$\text{Mode} < \text{Median} < \text{Mean}$$

► Negative Skew (Left-Skewed)

- Tail on the **left**
- Most values are large
- **Skewness** < 0

$$\text{Mean} < \text{Median} < \text{Mode}$$

► No Skew (Symmetric Distribution)

- Balanced distribution
- **Skewness** = 0

$$\text{Mean} = \text{Median} = \text{Mode}$$

Skewness Formula

For a sample:

$$\text{Skewness} = \frac{n}{(n-1)(n-2)} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{s} \right)^3$$

Where:

- $\bar{x}$ = sample mean
 - s = sample standard deviation
 - n = sample size
-

Summary Table

Skew Type	Tail Direction	Skewness	Mean vs Median vs Mode
Positive (Right)	Right tail	> 0	Mode < Median < Mean
Negative (Left)	Left tail	< 0	Mean < Median < Mode
No Skew (Symmetric)	Even	= 0	Mean = Median = Mode

Variance, Standard Deviation, Standard Error, Covariance, and Correlation

✓ Population vs. Sample Mean

Before we discuss variance, it's important to distinguish between:

- **Population Mean** (μ): The true mean of the entire population.
- **Sample Mean** ($\bar{x}$): The average of a sample drawn from the population.

We use the sample mean to estimate the population mean when the whole population is not accessible. This distinction affects how we calculate variance and other statistics.

✓ Variance

Concept:

- Variance measures the **spread** or **dispersion** of data around the mean.
 - It tells us **how far** individual values of a random variable deviate (on average) from the mean.
-

Formulas

Theoretical Variance:

$$\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$$

Population Variance:

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2$$

Sample Variance:

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

? Why Divide by $n - 1$ in Sample Variance?

When computing variance from a sample, we use the **sample mean** $\bar{x}$ as an estimate of the population mean μ . But this estimation "uses up" one degree of freedom — we have **less variability left to measure**.

- Using $\bar{x}$ makes the squared deviations $(x_i - \bar{x})^2$ **slightly smaller** than if we knew the true μ

- If we divide by n , we'd **underestimate** the true variance
- So we divide by $n - 1$ to correct this bias — this is known as **Bessel's correction**

By dividing by $n - 1$, the **average of sample variances** across many samples will match the **true population variance**.

✓ Standard Deviation

Concept:

- The square root of the variance.
 - Measures the **typical distance** from the mean.
 - **Preferred over variance** because it has the same **units** as the original data.
-

Formulas:

Population Standard Deviation:

$$\sigma = \sqrt{\sigma^2} = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2}$$

Sample Standard Deviation:

$$s = \sqrt{s^2} = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}$$

✓ Standard Error (SE)

Concept:

- Standard error measures the **uncertainty in the estimate** of a statistic (typically the mean).
 - It tells you **how much the sample mean ($\bar{x}$)** would vary **from sample to sample**.
-

Formula:

$$SE_{\bar{x}} = \frac{s}{\sqrt{n}}$$

Where:

- s : **Sample standard deviation**
 - n : **Sample size**
-

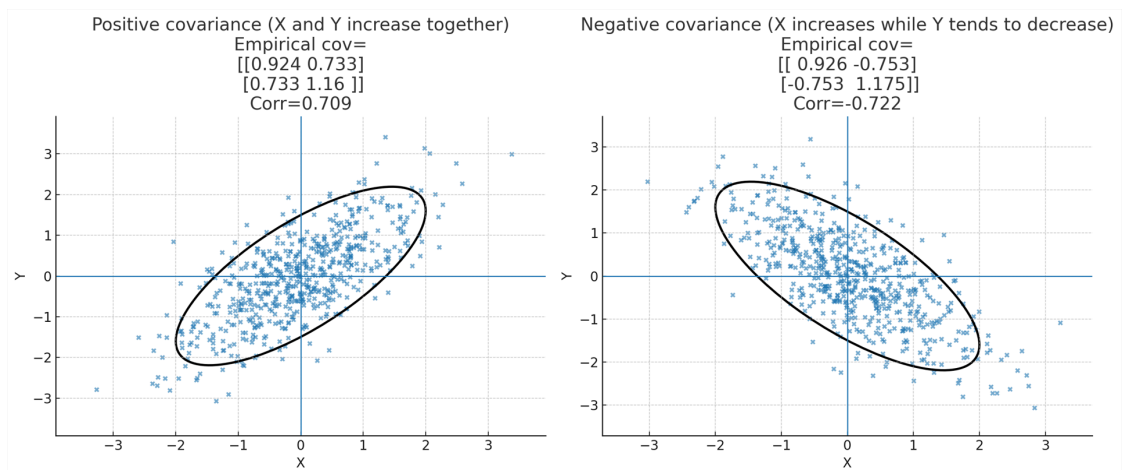
✓ Difference Between Standard Deviation and Standard Error

Feature	Standard Deviation (SD)	Standard Error (SE)
Definition	Spread of individual data points	Spread of sample statistic (mean)
Formula	$s = \sqrt{\frac{1}{n-1} \sum (x_i - \bar{x})^2}$	$SE = \frac{s}{\sqrt{n}}$
Usage	Describes variability in data	Describes precision of estimate (like mean)
Decreases with more data?	No	Yes, as n increases
Units	Same as data	Same as data

✓ Covariance

Concept:

- Measures the **direction** of the **linear relationship** between two variables.
- A positive value means both variables tend to increase together.



Formulas

Theoretical:

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])] = \mathbb{E}[XY] - \mathbb{E}[X] \cdot \mathbb{E}[Y]$$

Population Covariance:

$$\text{Cov}(X, Y) = \frac{1}{N} \sum_{i=1}^N (x_i - \mu_X)(y_i - \mu_Y)$$

Sample Covariance:

$$s_{xy} = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

Special Case: Covariance of the Same Variable

When we compute the covariance of a random variable with itself:

$$\text{Cov}(X, X) = \text{Var}(X)$$

So, covariance is a generalization of variance.

We know:

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])] = \mathbb{E}[XY] - \mathbb{E}[X] \cdot \mathbb{E}[Y]$$

Now, if we set $Y = X$:

$$\text{Cov}(X, X) = \mathbb{E}[(X - \mathbb{E}[X])(X - \mathbb{E}[X])]$$

which simplifies to:

$$\text{Cov}(X, X) = \mathbb{E}[(X - \mathbb{E}[X])^2] = \text{Var}(X)$$

✓ Hence, the covariance of a random variable with itself is exactly its variance.

Covariance Matrix

For multiple random variables (say $X_1, X_2, \dots, X_p$), we can arrange the covariances into a **covariance matrix**:

$$\Sigma = \begin{bmatrix} \text{Var}(X_1) & \text{Cov}(X_1, X_2) & \dots & \text{Cov}(X_1, X_p) \\ \text{Cov}(X_2, X_1) & \text{Var}(X_2) & \dots & \text{Cov}(X_2, X_p) \\ \vdots & \vdots & \ddots & \vdots \\ \text{Cov}(X_p, X_1) & \text{Cov}(X_p, X_2) & \dots & \text{Var}(X_p) \end{bmatrix}$$

- The **diagonal elements** are the variances ($\text{Var}(X_i)$).
 - The **off-diagonal elements** are covariances between different variables ($\text{Cov}(X_i, X_j)$).
-

Application: PCA (Principal Component Analysis)

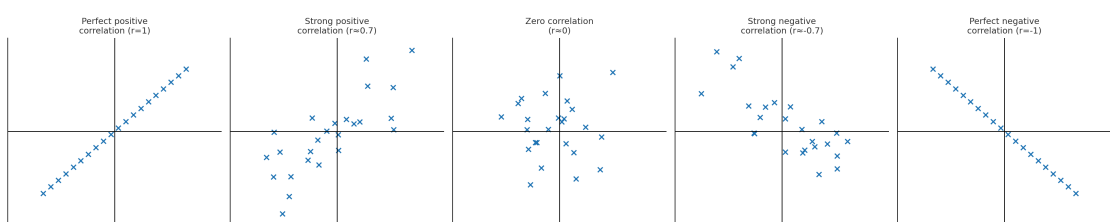
- PCA uses the **covariance matrix** of the data.
 - By finding the **eigenvectors and eigenvalues** of Σ , we identify the directions (principal components) that capture the **maximum variance** in the data.
 - The diagonal variances indicate how much spread each variable has, while the off-diagonal covariances show how variables move together.
 - PCA reduces dimensionality by projecting data onto the directions of largest variance.
 - The covariance matrix Σ is a **positive semi-definite (PSD) matrix**, meaning all its eigenvalues are **non-negative**.
-

✓ Correlation

Concept:

- A **scaled** version of covariance.
- Measures **strength and direction** of a **linear relationship**.
- Always lies in the range ([-1, 1]).

Correlation coefficient	Correlation strength	Correlation type
-0.7 to -1	Very strong	Negative
-0.5 to -0.7	Strong	Negative
-0.3 to -0.5	Moderate	Negative
0 to -0.3	Weak	Negative
0	None	Zero
0 to 0.3	Weak	Positive
0.3 to 0.5	Moderate	Positive
0.5 to 0.7	Strong	Positive
0.7 to 1	Very strong	Positive



Formulas

Theoretical (Pearson):

$$\rho_{X,Y} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)} \cdot \sqrt{\text{Var}(Y)}}$$

Sample (Pearson):

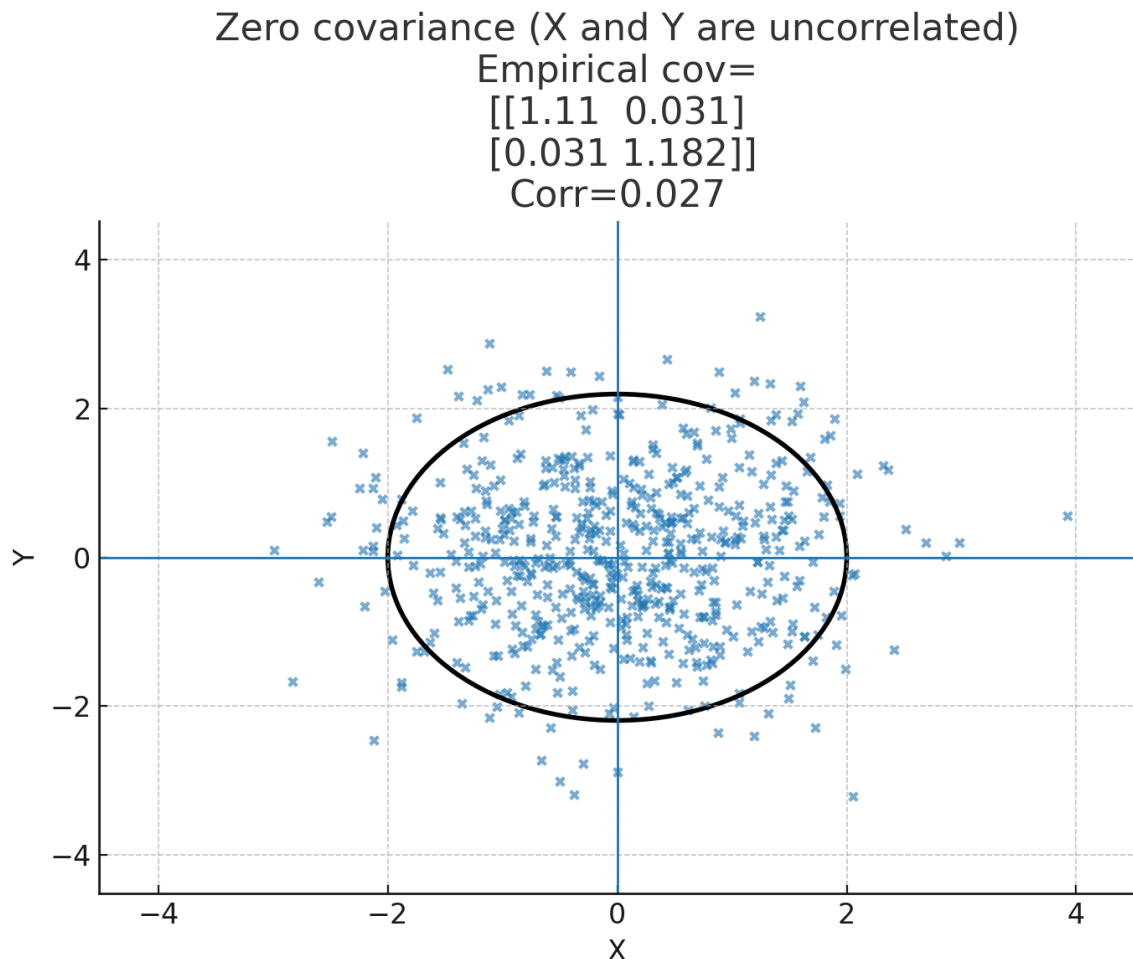
$$r = \frac{s_{xy}}{s_x \cdot s_y}$$

Where:

- s_{xy} : **Sample covariance**
- s_x, s_y : **Sample standard deviations**

Zero Covariance, Independence, and Correlation vs Causation

1. Zero covariance does **not** imply independence, but independence implies zero covariance



Definitions:

- Covariance measures *linear dependence* between random variables X and Y :

$$\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$$

- Independence means that knowing X gives no information about Y , and vice versa.

a) Independence implies zero covariance

If X and Y are independent, then:

$$\mathbb{E}[XY] = \mathbb{E}[X] \cdot \mathbb{E}[Y]$$

Therefore:

$$\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] = 0$$

b) Zero covariance does **NOT** imply independence

Counterexample:

Let

$$X \sim \mathcal{U}(-1, 1) \quad (\text{uniform distribution over } [-1, 1])$$

and define

$$Y = X^2$$

- X and Y are clearly dependent since Y is determined by X .
- However,

$$\mathbb{E}[X] = 0, \quad \mathbb{E}[Y] = \frac{1}{3}, \quad \mathbb{E}[XY] = \mathbb{E}[X^3] = 0$$

Thus,

$$\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] = 0 - (0)\left(\frac{1}{3}\right) = 0$$

Despite zero covariance, X and Y are *not independent*.

Key Insight

Covariance (and correlation, which is just the normalized covariance) only captures **linear dependence** between variables.

If X and Y follow a **nonlinear relationship** (like a parabola $Y = X^2$), the linear association averages out (positive and negative contributions cancel), so covariance = 0.

But X and Y are still strongly dependent — in fact, Y is completely determined by X !

2. Correlation does **not** imply causation

Correlation measures the *linear association* between two variables:

$$\rho_{X,Y} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

A high correlation does **not** mean one variable causes the other. There may be:

- A hidden common cause (confounder)
 - Reverse causality
 - Pure coincidence
-

Real-world example:

- Ice cream sales and drowning incidents tend to increase together (high correlation).
 - This does **not** mean ice cream sales cause drownings.
 - The hidden variable is **temperature (summer season)**, which increases both ice cream sales and swimming activity (leading to more drownings).
-

Summary:

Statement	True/False	Explanation
Independence $\Rightarrow$ Zero covariance	True	Independence removes all dependence
Zero covariance $\Rightarrow$ Independence	False	May have nonlinear dependence
Correlation $\Rightarrow$ Causation	False	Correlation alone does not prove cause



Correlation vs Coefficient of Determination



What is R (Correlation Coefficient)?

- Measures the **strength and direction** of a **linear relationship** between observed and predicted values.

$$R \in [-1, 1]$$

- $R = 1$: perfect positive linear
- $R = 0$: no linear relationship
- $R = -1$: perfect negative linear



What is R^2 (Coefficient of Determination)?

- Measures the **proportion of variance in Y** explained by the model.
- Indicates **goodness of fit**.

$$R^2 = 1 - \frac{SS_{\text{res}}}{SS_{\text{tot}}}$$

Where:

- $SS_{\text{res}} = \sum (y_i - \hat{y}_i)^2$
- $SS_{\text{tot}} = \sum (y_i - \bar{y})^2$



In Simple Linear Regression:

$$R^2 = (R_{\text{Pearson}})^2$$



Intuition

Metric	Meaning
R	Strength & direction of linear relationship
R^2	Fraction of variance in Y explained by X



Comparison Table

Feature	R (Correlation)	R^2 (Goodness of Fit)
Measures direction?	✓ Yes	✗ No
Range	$[-1, 1]$	$[0, 1]$
Interprets strength?	✓ Yes	✓ Yes (proportion explained)
Useful for	Linearity, relationship	Regression model evaluation
Shows model fit?	✗ Not directly	✓ Yes
Affected by outliers?	✓ Yes	✓ Yes



Summary

- Use R to check **relationship strength and direction**.
- Use R^2 to evaluate **how well your model explains the data**.

How to Calculate and Compare Correlation Coefficients

1. Pearson Correlation (r)

Concept:

- Measures **linear relationship** between two continuous variables.
- **Parametric test** → Assumes data is approximately **normally distributed** and variables have equal variance.

Formula:

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \cdot \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}}$$

- $r \in [-1, 1]$
-

2. Spearman's Rank Correlation (ρ_s)

Concept:

- Measures **monotonic relationship** (not necessarily linear)
- **Non-parametric** → Does **not** require normality.
- Based on **ranks** of the data.

Formula (if no ties):

$$\rho_s = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}$$

Where:

- d_i = difference between ranks of x_i and y_i

If there are ties:

- Replace raw data with **ranks** (average ranks for ties)
- Then compute **Pearson correlation** on those ranks:

$$\rho_s = r_{\text{Pearson on ranks}}$$

Example: Two Ways to Compute ρ_s

Data:

Observation	X	Y
1	10	8
2	20	12
3	20	15
4	40	20
5	50	25

Step 1 — Assign ranks (average for ties)

Obs	X	Rank(X)	Y	Rank(Y)
1	10	1	8	1
2	20	2.5	12	2
3	20	2.5	15	3
4	40	4	20	4
5	50	5	25	5

Method 1 — Original Spearman Formula

$$d_i = \text{Rank}(X) - \text{Rank}(Y)$$

Obs	Rank(X)	Rank(Y)	d_i	d_i^2
1	1	1	0	0
2	2.5	2	0.5	0.25
3	2.5	3	-0.5	0.25
4	4	4	0	0
5	5	5	0	0

$$\sum d_i^2 = 0.5$$

$$\rho_s = 1 - \frac{6(0.5)}{5(5^2 - 1)} = 1 - \frac{3}{120} = 0.975$$

Method 2 — Pearson Correlation on Ranks

Pearson's r computed using Rank(X) and Rank(Y) also gives:

$$\rho_s = 0.975$$

✓ 3. Kendall's Tau (τ)

Concept:

- Measures **ordinal association** by comparing all pairs of observations.
- Non-parametric** → Does **not** require normality.
- Works better than ρ_s when there are **many tied ranks**.

Formula:

$$\tau = \frac{C - D}{\frac{1}{2}n(n - 1)}$$

Where:

- C = number of **concordant pairs**
- D = number of **discordant pairs**

◆ Example: Kendall's Tau

Data:

Observation	X	Y
1	12	30
2	20	50

Observation	X	Y
3	28	40

Step 1 — List all pairs (i, j) where $i < j$:

- Pairs: (1,2), (1,3), (2,3)

Step 2 — Check each pair:

- (1,2): X increases, Y increases → **Concordant**
- (1,3): X increases, Y increases → **Concordant**
- (2,3): X increases, Y decreases → **Discordant**

Step 3 — Count:

- $C = 2, D = 1, n = 3$

Step 4 — Apply formula:

$$\tau = \frac{2 - 1}{\frac{1}{2} \cdot 3 \cdot (3 - 1)} = \frac{1}{3} \approx 0.333$$

Comparison Table

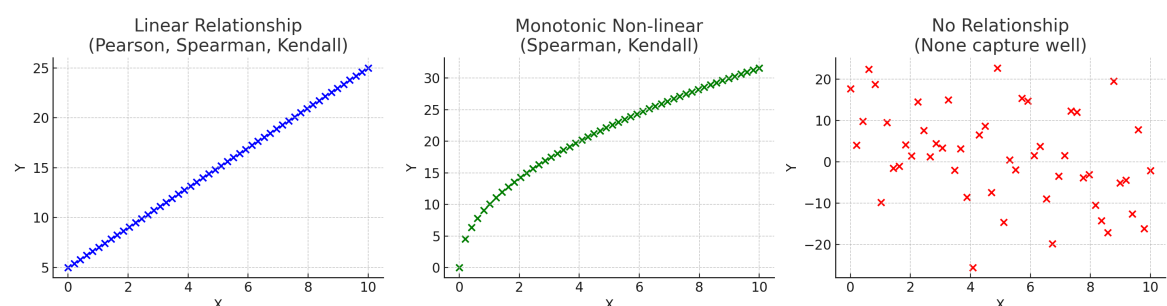
Feature	Pearson (r)	Spearman (ρ_s)	Kendall (τ)
Type of Relationship	Linear	Monotonic (rank-based)	Ordinal (pairwise concordance)
Data Type (X, Y)	Interval/ratio (metric)	Ordinal, interval/ratio	Ordinal, interval/ratio
Based On	Raw values	Ranks	Concordant vs discordant pairs
Sensitive to Outliers	✓ High	✗ Low	✗ Very low
Parametric?	✓ Yes	✗ No (non-parametric)	✗ No (non-parametric)
Handles Many Ties Well	✗ No	⚠ Moderate	✓ Yes
Works with Linear Data	✓ Yes	✓ Yes	✓ Yes
Works with Non-linear Monotonic Data	✗ No	✓ Yes	✓ Yes
Typical Use Cases	Normal numeric data	Skewed, ordinal, non-linear	Ordinal data, many ties, small n

Data Types for Correlation

All three correlation measures typically require **metric** data for the variables (unless purely ordinal in Kendall's case).

Metric data examples:

- **Interval scale:** Temperature in °C, IQ scores
- **Ratio scale:** Weight (kg), height (cm), reaction time (ms), income (\$)





Discrete vs Continuous Random Variables



Discrete Random Variable (DRV)

- Takes **countable** values (finite or countably infinite)
 - Gaps between values
 - Examples:
 - Number of emails per day
 - Number of students in a class
 - Dice rolls
-



Continuous Random Variable (CRV)

- Takes values from a **real interval**
 - **Uncountably infinite** possibilities
 - Examples:
 - Height, weight, time, temperature
-



PMF vs PDF vs CDF

◆ 1. PMF (Probability Mass Function)

- **Used for discrete random variables**
- Gives probability of exact value

$$P(X = x) = p(x), \quad \sum_x p(x) = 1$$

Example (fair die):

$$p(x) = \frac{1}{6}, \quad x \in \{1, 2, 3, 4, 5, 6\}$$

◆ 2. PDF (Probability Density Function)

- **Used for continuous random variables**
- Not actual probability at a point, but **density**
- Area under curve gives probability

$$P(a \leq X \leq b) = \int_a^b f(x) dx$$

- Total area:

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

- Note:

$$P(X = x) = 0 \quad \text{for any } x$$

◆ 3. CDF (Cumulative Distribution Function)

- Used for both **discrete** and **continuous**
- Probability that X is less than or equal to a value:

$$F(x) = P(X \leq x)$$

- For discrete:

$$F(x) = \sum_{t \leq x} p(t)$$

- For continuous:

$$F(x) = \int_{-\infty}^x f(t) dt$$

Relationship Between PDF/PMF and CDF

1. Discrete Distributions

- The **PMF** gives the probability of exact outcomes:

$$P(X = x) = f(x)$$

- The **CDF** is the cumulative sum of the PMF:

$$F(x) = P(X \leq x) = \sum_{t \leq x} f(t)$$

- To recover the PMF from the CDF:

$$f(x) = F(x) - F(x - 1)$$

2. Continuous Distributions

- The **PDF** is the derivative of the CDF:

$$f(x) = \frac{d}{dx} F(x)$$

- The **CDF** is the integral of the PDF:

$$F(x) = \int_{-\infty}^x f(t) dt$$

Properties:

- $F(-\infty) = 0$
 - $F(\infty) = 1$
 - Non-decreasing
-

Joint Probability & Joint Conditional Probability in random variables notations

◆ Discrete Case

Joint Probability (Discrete):

$$P(X = x, Y = y) = P(X = x \cap Y = y)$$

Joint Conditional Probability (Discrete):

$$P(X = x, Y = y \mid Z = z) = \frac{P(X = x, Y = y, Z = z)}{P(Z = z)} \quad \text{for } P(Z = z) > 0$$

Chain Rule (Discrete):

$$P(X = x, Y = y) = P(X = x \mid Y = y) \cdot P(Y = y)$$

$$P(X = x, Y = y \mid Z = z) = P(X = x \mid Y = y, Z = z) \cdot P(Y = y \mid Z = z)$$

Marginalization (Discrete):

$$P(X = x) = \sum_y P(X = x, Y = y)$$

◆ Continuous Case

Joint Probability Density Function (PDF):

$$f_{X,Y}(x, y) = \text{Joint PDF of } X \text{ and } Y$$

Joint Conditional PDF:

$$f_{X,Y|Z}(x, y \mid z) = \frac{f_{X,Y,Z}(x, y, z)}{f_Z(z)} \quad \text{for } f_Z(z) > 0$$

Chain Rule (Continuous):

$$f_{X,Y}(x, y) = f_{X|Y}(x \mid y) \cdot f_Y(y)$$

$$f_{X,Y|Z}(x, y \mid z) = f_{X|Y,Z}(x \mid y, z) \cdot f_{Y|Z}(y \mid z)$$

Marginalization (Continuous):

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy$$

📌 Expected Value and Variance: Concepts and Properties

🧠 What is Expected Value?

- Represents the **long-run average** outcome of a random variable.
- Also called the **mean** or **expectation**.
- Think of it as the "**center of mass**" or **weighted average** of a random variable.

◆ Discrete Case:

$$\mathbb{E}[X] = \sum_x x \cdot p(x)$$

◆ Continuous Case:

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x \cdot f(x) dx$$

💰 Example: Financial Gain/Loss Game

Outcome	Amount (x)	Probability $p(x)$
Win big	+\$100	0.1

Outcome	Amount (x)	Probability $p(x)$
Win small	+\$20	0.3
Lose small	-\$10	0.4
Lose big	-\$50	0.2

$$\mathbb{E}[X] = (100)(0.1) + (20)(0.3) + (-10)(0.4) + (-50)(0.2) = 10 + 6 - 4 - 10 = 2$$

Interpretation: You expect to **gain \$2 per game** in the long run.

Expected Value of a Function: $\mathbb{E}[g(X)]$

Discrete:

$$\mathbb{E}[g(X)] = \sum_x g(x) \cdot p(x)$$

Continuous:

$$\mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x) \cdot f(x) dx$$

Linearity of Expectation

Always holds, even without independence.

For constants a, b :

$$\mathbb{E}[aX + b] = a \cdot \mathbb{E}[X] + b$$

For two random variables X and Y :

$$\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$$

Variance Transformations

Variance Definition:

$$\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2]$$

◆ Add/Subtract Constant:

$$\text{Var}(X + c) = \text{Var}(X)$$

◆ Multiply by Constant:

$$\text{Var}(aX) = a^2 \cdot \text{Var}(X)$$

The Convolution Theorem for Sums of Random Variables

Discrete Random Variables

Let $Y = X_1 + X_2$, where X_1 and X_2 are independent discrete random variables with pmfs $p_{X_1}(k)$ and $p_{X_2}(k)$.

The pmf of Y is given by the **convolution sum**:

$$p_Y(j) = \sum_k p_{X_1}(k) p_{X_2}(j - k)$$

for all values of j .

Continuous Random Variables

If X_1 and X_2 have pdfs $p_1(x_1)$ and $p_2(x_2)$, the cumulative distribution function (cdf) of $Y = X_1 + X_2$ is:

$$P_Y(y) = P(Y \leq y) = \iint_{x_1 + x_2 \leq y} p_1(x_1) p_2(x_2) dx_1 dx_2$$

Differentiating under the integral sign gives the pdf of Y :

$$p_Y(y) = \frac{d}{dy} P_Y(y) = \int p_1(x_1) p_2(y - x_1) dx_1$$

This integral is the **convolution** of p_1 and p_2 :

$$p_Y(y) = (p_1 * p_2)(y)$$

Intuition

- The convolution sums/integrates over all ways to split y into components x_1 and $y - x_1$.
 - For discrete variables, it's a summation; for continuous variables, an integral.
 - Convolution is like "flipping and sliding" one function over the other.
-

Example: Sum of Two Dice Rolls

Each die roll $X_i \sim \text{Uniform}\{1, 2, \dots, 6\}$, with

$$p_{X_i}(k) = \frac{1}{6}, \quad k = 1, 2, \dots, 6$$

Then

$$p_Y(2) = p_{X_1}(1) \cdot p_{X_2}(1) = \frac{1}{6} \cdot \frac{1}{6} = \frac{1}{36}$$

$$p_Y(3) = p_{X_1}(1) \cdot p_{X_2}(2) + p_{X_1}(2) \cdot p_{X_2}(1) = \frac{1}{6} \cdot \frac{1}{6} + \frac{1}{6} \cdot \frac{1}{6} = \frac{2}{36}$$

and so on, with probabilities forming a distribution that resembles a Gaussian shape.

Example: Sum of Two Gaussian Variables

If

$$X_1 \sim \mathcal{N}(\mu_1, \sigma_1^2), \quad X_2 \sim \mathcal{N}(\mu_2, \sigma_2^2)$$

then

$$Y = X_1 + X_2 \sim \mathcal{N}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$$

The convolution of two Gaussian pdfs is another Gaussian pdf with summed means and variances. **This result is known as the *Convolution Theorem* for sums of independent random variables.**

Central Limit Theorem (CLT) –

Conceptual Understanding

The Central Limit Theorem (CLT) states that:

If we take sufficiently large random samples from any population with a finite mean and variance, the distribution of the sample means will tend to follow a normal distribution (bell-shaped), regardless of the shape of the original distribution.

This theorem is **fundamental in statistics**, as it justifies why the **normal distribution appears so often** in practice.

Mathematical Formulation

Let $X_1, X_2, \dots, X_n$ be **independent and identically distributed (i.i.d.)** random variables from a population with:

- Mean: $\mu = \mathbb{E}[X_i]$
- Variance: $\sigma^2 = \text{Var}(X_i)$

Define the **sample mean** as:

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$$

Then, the Central Limit Theorem states:

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} \mathcal{N}(0, 1) \quad \text{as } n \rightarrow \infty$$

Where:

- $\bar{X}_n$ is the sample mean,
 - μ is the population mean,
 - σ is the population standard deviation,
 - $\mathcal{N}(0, 1)$ is the **standard normal distribution**,
 - $\xrightarrow{d}$ means **converges in distribution**.
-

Example: Sampling from a Uniform Distribution

Suppose we sample from a **Uniform(0, 1)** distribution. We know:

- $\mu = \frac{0+1}{2} = 0.5$
- $\sigma^2 = \frac{(1-0)^2}{12} = \frac{1}{12}$
- $\sigma = \frac{1}{\sqrt{12}} \approx 0.2887$
- Sample size: $n = 32$

The standard deviation of the sample mean is:

$$\text{SD}(\bar{X}_{32}) = \frac{\sigma}{\sqrt{n}} = \frac{1/\sqrt{12}}{\sqrt{32}} = \frac{1}{\sqrt{384}} \approx 0.051$$

So, the standardized sample mean becomes:

$$Z = \frac{\bar{X}_{32} - 0.5}{1/\sqrt{384}} \approx \frac{\bar{X}_{32} - 0.5}{0.051}$$

This Z will follow approximately $\mathcal{N}(0, 1)$ due to the CLT.

Practical Notes

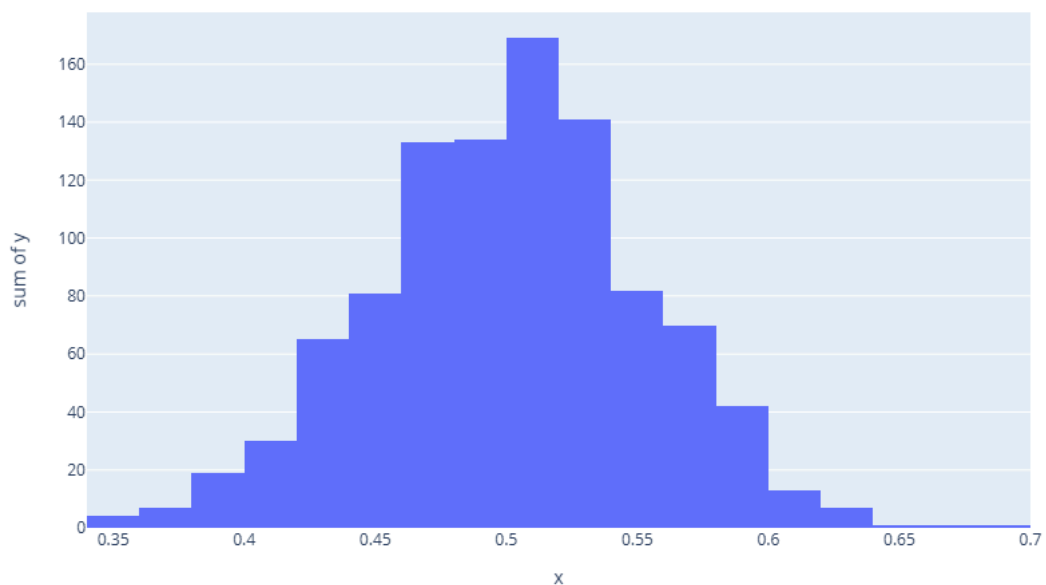
- A **sample size of $n \geq 30$** is usually sufficient for the CLT to hold — especially if the original distribution is symmetric (like uniform).
- If the distribution is **very skewed**, larger samples (e.g., $n \geq 50$ or 100) may be required.
- To visualize the normal shape of the sample mean distribution, you should **repeat the sampling process** (i.e., draw new samples and compute their means) at least **1000 times**.

Important Note on Assumptions

The CLT assumes that the **original population distribution** has a **finite mean and finite variance**:

- That is, $\mu = \mathbb{E}[X_i]$ and $\sigma^2 = \text{Var}(X_i)$ must **exist and be finite**.
- These refer to the **true properties of the population, not the sample**.
- If the original distribution has **infinite variance** (e.g., Cauchy distribution), then the CLT **does not apply**, and the sample mean may not converge to a normal distribution.

the Central Limit Theorem by repeatedly (1,000 times) taking samples of size 31 from a uniform distribution between 0 and 1, computing each sample's mean, and then plotting a histogram of these means — which forms an approximately normal distribution despite the original data being uniform.

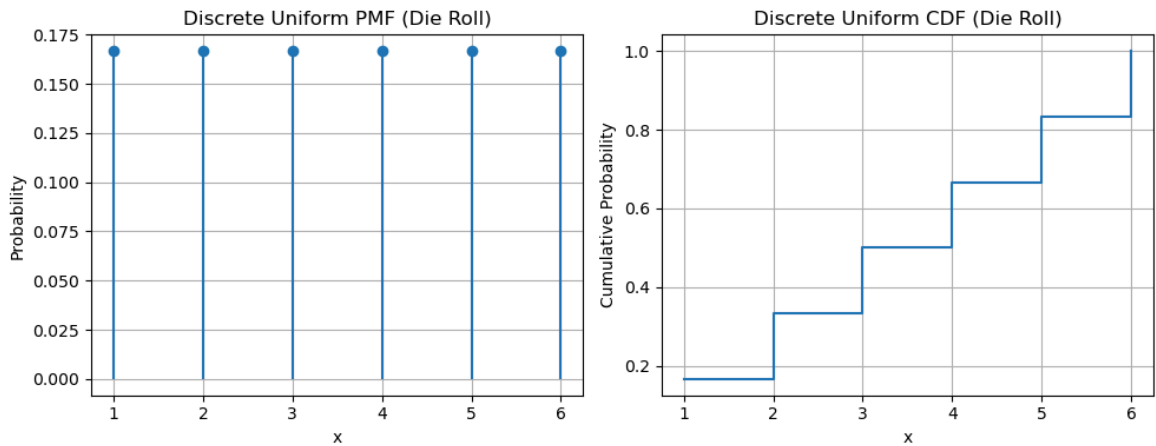




Discrete Probability Distributions



Discrete Uniform Distribution



Discrete Uniform Distribution

Concept & Intuition:

A discrete uniform distribution assigns equal probability to all integers between a minimum value a and a maximum value b . Every outcome is equally likely.

Example:

Rolling a fair six-sided die where $X \in \{1, 2, 3, 4, 5, 6\}$

Parameters:

- a : minimum integer value
- b : maximum integer value

PMF:

$$P(X = x) = \begin{cases} \frac{1}{b - a + 1}, & x \in \{a, a + 1, \dots, b\}, \\ 0, & \text{otherwise.} \end{cases}$$

CDF:

$$F(x) = \begin{cases} 0, & x < a, \\ \frac{\lfloor x \rfloor - a + 1}{b - a + 1}, & a \leq x \leq b, \\ 1, & x > b. \end{cases}$$

Mean:

$$\mathbb{E}[X] = \frac{a + b}{2}$$

Variance:

$$\text{Var}(X) = \frac{(b - a + 1)^2 - 1}{12}$$

✓ Worked Example (Die Roll)

Question:

Suppose you roll a fair six-sided die. Let X be the outcome, so $X \in \{1, 2, 3, 4, 5, 6\}$.

1. What is the probability of rolling a 4?
2. What is the probability that the outcome is at most 4 (i.e., $P(X \leq 4)$)?
3. What are the expected value and variance of X ?

Solution:

- PMF (single value):

$$P(X = 4) = \frac{1}{6}$$

- CDF (up to 4):

$$P(X \leq 4) = \frac{4 - 1 + 1}{6} = \frac{4}{6} = \frac{2}{3}$$

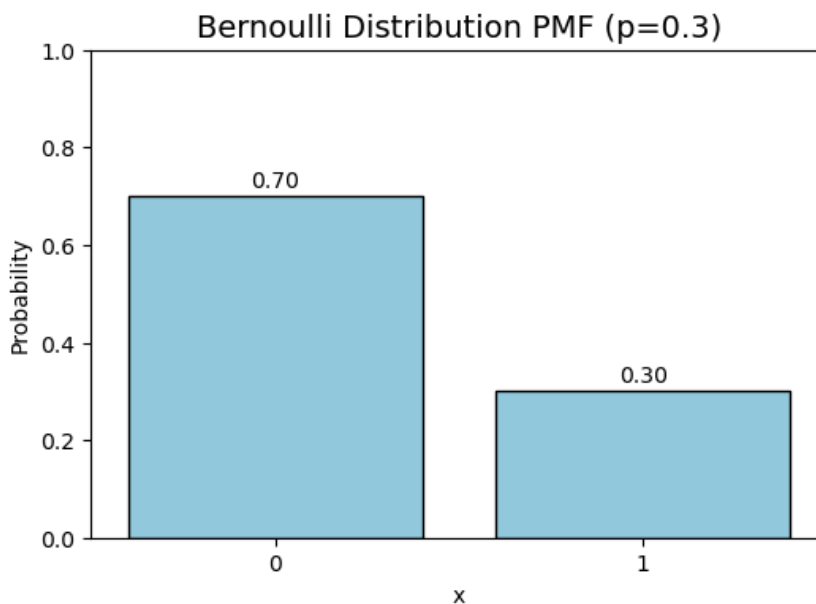
- Mean:

$$\mathbb{E}[X] = \frac{1 + 6}{2} = 3.5$$

- Variance:

$$\text{Var}(X) = \frac{(6 - 1 + 1)^2 - 1}{12} = \frac{35}{12} \approx 2.92$$

✓ Bernoulli Distribution



Concept & Intuition:

The Bernoulli distribution models a single trial that results in either success (1) with probability p , or failure (0) with probability $1 - p$. It is the simplest discrete distribution and the basis for more complex ones like the Binomial.

Example:

Flipping a biased coin with $P(\text{heads}) = 0.7$ (success)

Parameters:

- p : probability of success ($0 \leq p \leq 1$)

PMF:

$$P(X = x) = p^x(1 - p)^{1-x}, \quad x \in \{0, 1\}$$

Mean:

$$\mathbb{E}[X] = p$$

Variance:

$$\text{Var}(X) = p(1 - p)$$

Example Plugged In:

Let $p = 0.7$

- PMF:

$$P(X = 1) = 0.7, \quad P(X = 0) = 0.3$$

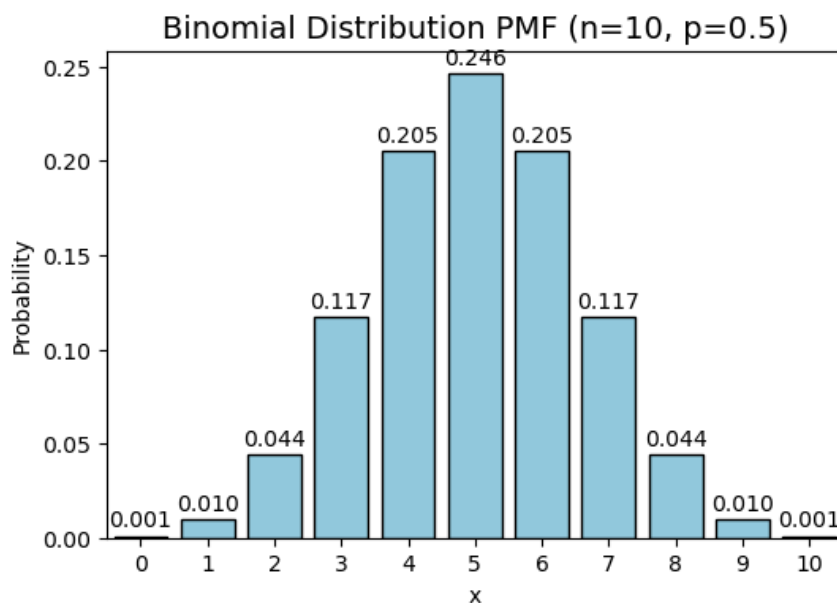
- Mean:

$$\mathbb{E}[X] = 0.7$$

- Variance:

$$\text{Var}(X) = 0.7 \times 0.3 = 0.21$$

Binomial Distribution



Concept & Intuition:

The Binomial distribution models the number of successes in a fixed number of independent Bernoulli trials. Each trial has the same probability p of success. It's used when you repeat an experiment a number of times and count how many times it succeeds.

Example:

Flipping a biased coin $n = 5$ times, with $P(\text{heads}) = 0.6$.

Parameters:

- n : number of independent trials

- p : probability of success in each trial

PMF:

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k = 0, 1, \dots, n$$

Mean:

$$\mathbb{E}[X] = np$$

Variance:

$$\text{Var}(X) = np(1-p)$$

Example Plugged In:

Let $n = 5, p = 0.6, k = 2$

- PMF:

$$P(X = 2) = \binom{5}{2} (0.6)^2 (0.4)^3 = 10 \cdot 0.36 \cdot 0.064 = 0.2304$$

- Mean:

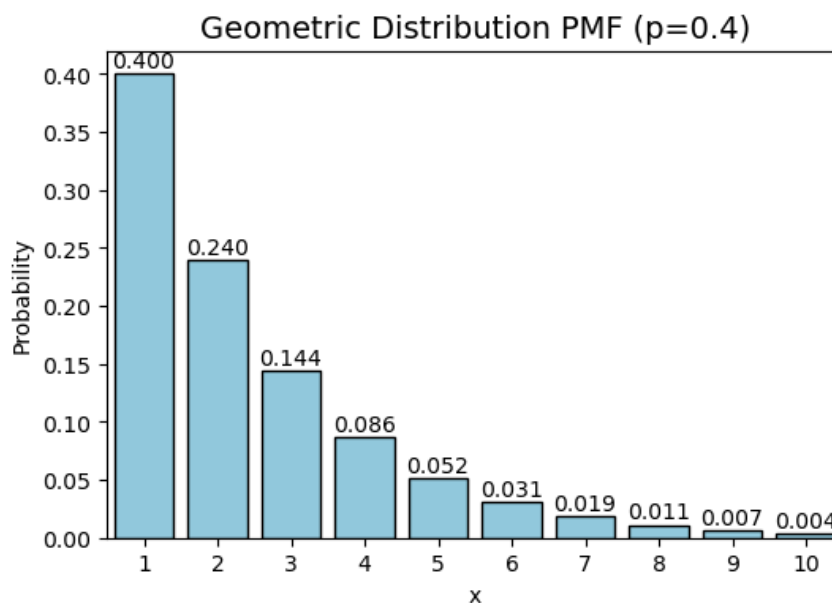
$$\mathbb{E}[X] = 5 \cdot 0.6 = 3$$

- Variance:

$$\text{Var}(X) = 5 \cdot 0.6 \cdot 0.4 = 1.2$$



Geometric Distribution



Concept & Intuition:

The Geometric distribution models the number of trials needed to get the **first success** in a sequence of independent Bernoulli trials with constant success probability p . It is unique among discrete distributions because it has the **memoryless property**.

Example:

Flipping a coin with $p = 0.25$ until the first head appears.

Parameters:

- p : probability of success on each trial

PMF:

$$P(X = k) = (1 - p)^{k-1}p, \quad k = 1, 2, 3, \dots$$

Mean:

$$\mathbb{E}[X] = \frac{1}{p}$$

Variance:

$$\text{Var}(X) = \frac{1 - p}{p^2}$$

Memoryless Property:

The geometric distribution is the **only discrete distribution** that satisfies:

$$P(X > s + t \mid X > s) = P(X > t)$$

Example Plugged In:

Let $p = 0.25$, $k = 3$

- PMF:

$$P(X = 3) = (0.75)^2 \cdot 0.25 = 0.1406$$

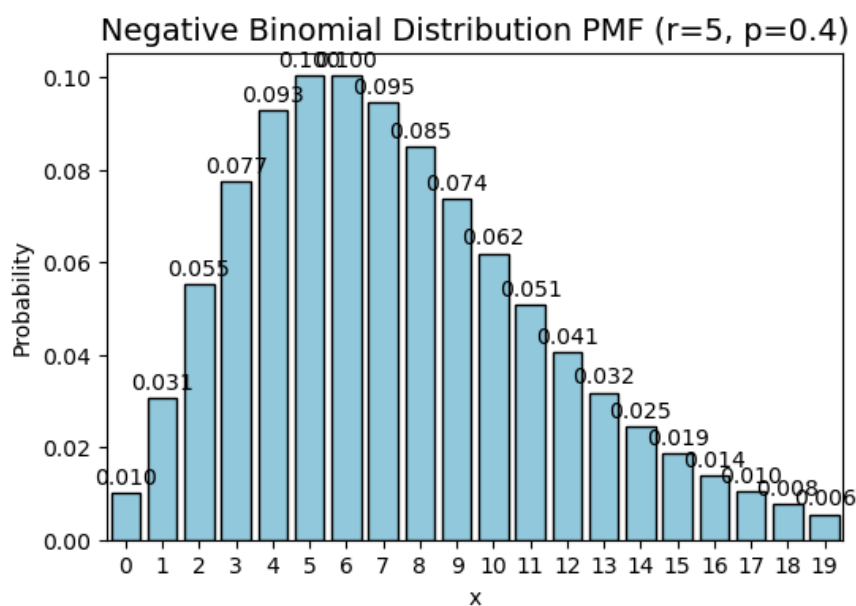
- Mean:

$$\mathbb{E}[X] = \frac{1}{0.25} = 4$$

- Variance:

$$\text{Var}(X) = \frac{1 - 0.25}{0.25^2} = \frac{0.75}{0.0625} = 12$$

Negative Binomial Distribution



Concept & Intuition:

The Negative Binomial distribution models the number of trials needed to achieve **r successes** in a series of

independent Bernoulli trials with success probability p . It generalizes the geometric distribution, which is the special case when $r = 1$.

Example:

How many times must we roll a die until we get 3 sixes?

Let $r = 3$, $p = \frac{1}{6}$ (probability of rolling a six)

Parameters:

- r : number of required successes
- p : probability of success on each trial

PMF:

$$P(X = k) = \binom{k-1}{r-1} p^r (1-p)^{k-r}, \quad k = r, r+1, r+2, \dots$$

Mean:

$$\mathbb{E}[X] = \frac{r}{p}$$

Variance:

$$\text{Var}(X) = \frac{r(1-p)}{p^2}$$

Example Plugged In:

Let $r = 3$, $p = \frac{1}{6}$, $k = 8$

- PMF:

$$P(X = 8) = \binom{7}{2} \left(\frac{1}{6}\right)^3 \left(\frac{5}{6}\right)^5 = 21 \cdot \frac{1}{216} \cdot \frac{3125}{7776} \approx 0.0196$$

- Mean:

$$\mathbb{E}[X] = \frac{3}{1/6} = 18$$

- Variance:

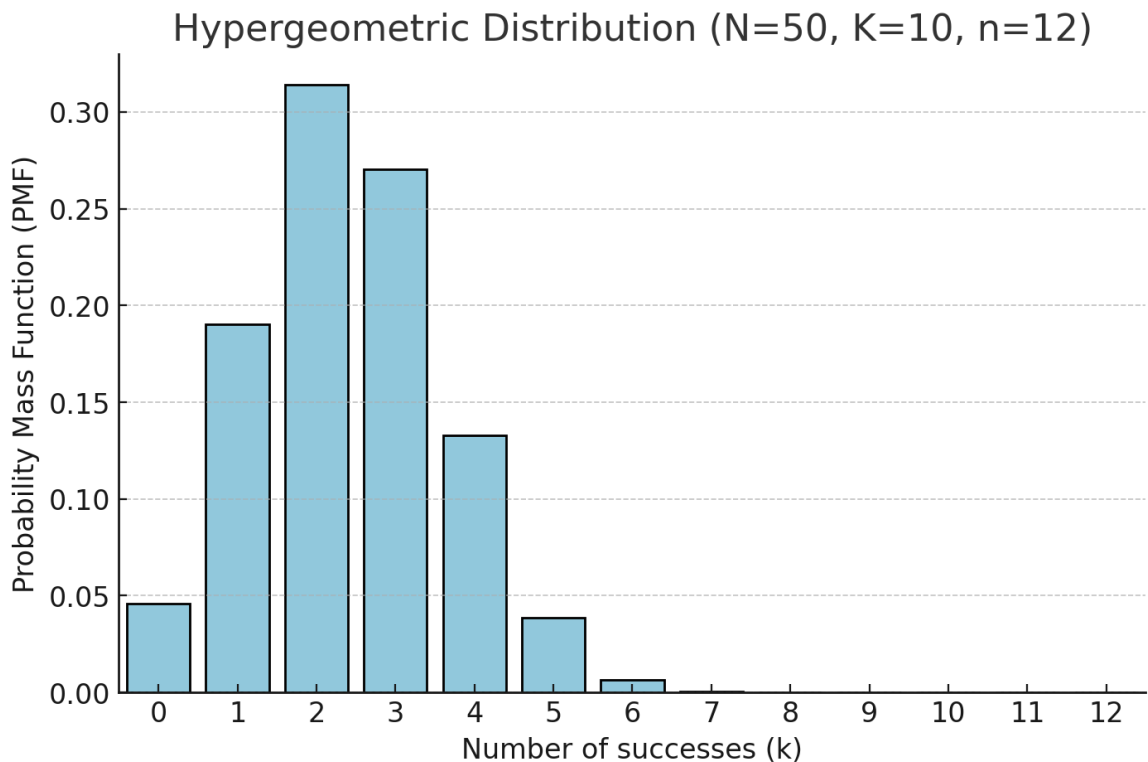
$$\text{Var}(X) = \frac{3 \cdot \frac{5}{6}}{(1/6)^2} = \frac{2.5}{1/36} = 90$$

Comparison: Binomial vs Geometric vs Negative Binomial

Property	Binomial	Geometric	Negative Binomial
What it counts	Number of successes in n trials	Number of trials until 1st success	Number of trials until r successes
Fixed	Number of trials (n)	Number of successes = 1	Number of successes (r)
Random	Number of successes	Number of trials	Number of trials
PMF	$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$	$P(X = k) = (1-p)^{k-1} p$	$P(X = k) = \binom{k-1}{r-1} p^r (1-p)^{k-r}$
Support	$k = 0, 1, \dots, n$	$k = 1, 2, 3, \dots$	$k = r, r+1, r+2, \dots$
Mean	$\mathbb{E}[X] = np$	$\mathbb{E}[X] = \frac{1}{p}$	$\mathbb{E}[X] = \frac{r}{p}$

Property	Binomial	Geometric	Negative Binomial
Variance	$\text{Var}(X) = np(1-p)$	$\text{Var}(X) = \frac{1-p}{p^2}$	$\text{Var}(X) = \frac{r(1-p)}{p^2}$
Memoryless Property	✗	✓	✗
Typical Use Case	Count successes in fixed trials	Time until first success	Time until r-th success

Hypergeometric Distribution



Concept & Intuition:

The Hypergeometric distribution models the number of successes in a **sample drawn without replacement** from a finite population that contains a certain number of successes. Unlike the Binomial, the trials are **not independent**, because we don't replace the items.

Example:

An urn contains $N = 20$ balls: $K = 7$ red (successes) and $N - K = 13$ blue (failures). You randomly draw $n = 5$ balls **without replacement**. Let X be the number of red balls drawn.

Parameters:

- N : total population size
- K : number of successes in the population
- n : sample size (number of draws)

PMF:

$$P(X = k) = \frac{\binom{K}{k} \binom{N-K}{n-k}}{\binom{N}{n}}, \quad \max(0, n - N + K) \leq k \leq \min(n, K)$$

Mean:

$$\mathbb{E}[X] = n \cdot \frac{K}{N}$$

Variance:

$$\text{Var}(X) = n \cdot \frac{K}{N} \cdot \frac{N-K}{N} \cdot \frac{N-n}{N-1}$$

Example Plugged In:

Let $N = 20$, $K = 7$, $n = 5$, and $k = 2$

- PMF:

$$P(X = 2) = \frac{\binom{7}{2} \binom{13}{3}}{\binom{20}{5}} = \frac{21 \cdot 286}{15504} \approx 0.3875$$

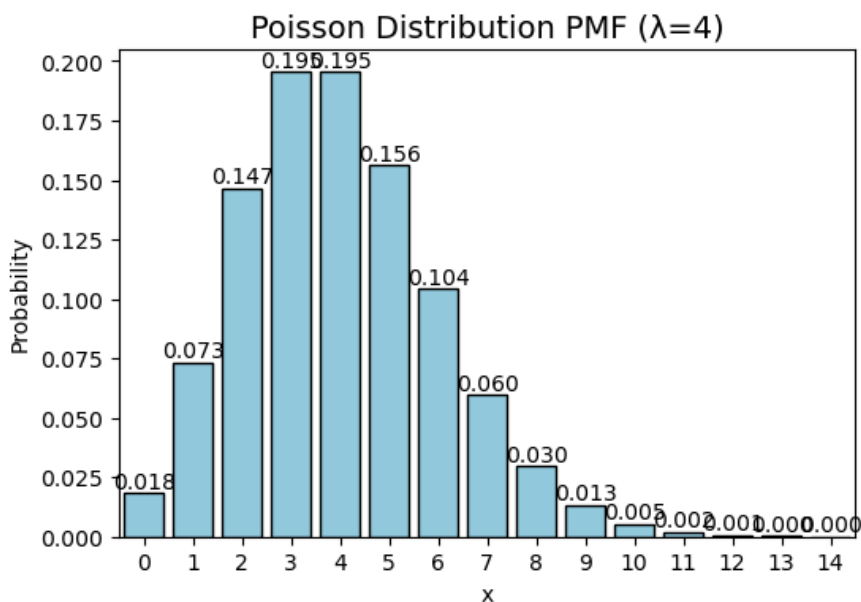
- Mean:

$$\mathbb{E}[X] = 5 \cdot \frac{7}{20} = 1.75$$

- Variance:

$$\text{Var}(X) = 5 \cdot \frac{7}{20} \cdot \frac{13}{20} \cdot \frac{15}{19} \approx 1.196$$

Poisson Distribution



Concept & Intuition:

The Poisson distribution models the number of events occurring in a **fixed interval of time or space**, assuming the events happen **independently** and at a **constant average rate** λ . It is ideal for modeling rare or random events.

It also arises as a **limiting case** of the Binomial distribution when $n \rightarrow \infty$, $p \rightarrow 0$, but $np = \lambda$ remains fixed.

Example:

A call center receives an average of $\lambda = 3$ calls per hour. Let X be the number of calls received in one hour.

Parameter:

- λ : expected number of events per interval ($\lambda > 0$)

PMF:

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}, \quad k = 0, 1, 2, \dots$$

Mean:

$$\mathbb{E}[X] = \lambda$$

Variance:

$$\text{Var}(X) = \lambda$$

Example Plugged In:

Let $\lambda = 3$, $k = 2$

- PMF:

$$P(X = 2) = \frac{3^2 \cdot e^{-3}}{2!} = \frac{9 \cdot e^{-3}}{2} \approx \frac{9 \cdot 0.0498}{2} \approx 0.224$$

- Mean:

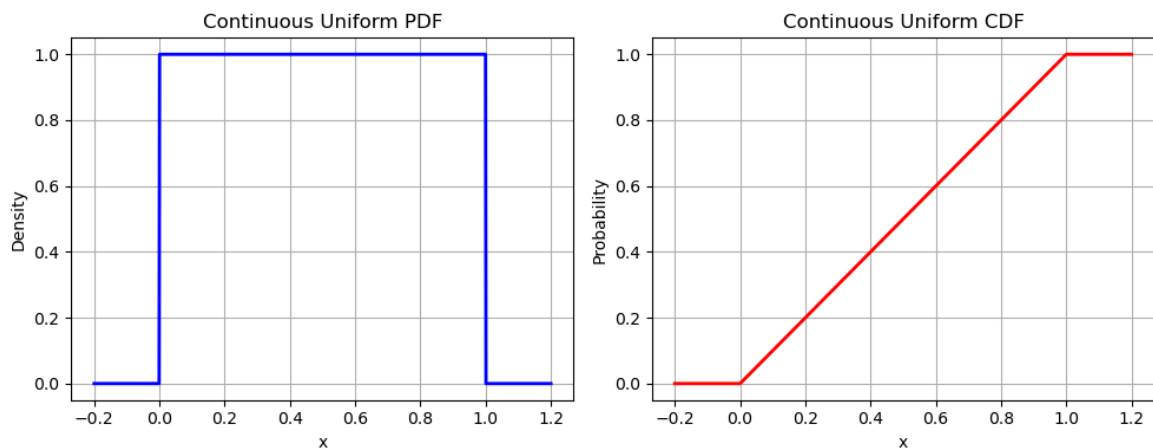
$$\mathbb{E}[X] = 3$$

- Variance:

$$\text{Var}(X) = 3$$

Continuous Probability Distributions

Continuous Uniform Distribution



Continuous Uniform Distribution

Concept & Intuition:

A continuous uniform distribution assigns equal probability density to all values in the interval $[a, b]$. Any point in this interval is equally likely.

Example:

Selecting a random number between 0 and 10 where $X \in [0, 10]$.

Parameters:

- a : minimum value

- b : maximum value

PDF:

$$f(x) = \begin{cases} \frac{1}{b-a}, & a \leq x \leq b, \\ 0, & \text{otherwise.} \end{cases}$$

CDF:

$$F(x) = \begin{cases} 0, & x < a, \\ \frac{x-a}{b-a}, & a \leq x \leq b, \\ 1, & x > b. \end{cases}$$

Mean:

$$\mathbb{E}[X] = \frac{a+b}{2}$$

Variance:

$$\text{Var}(X) = \frac{(b-a)^2}{12}$$



Worked Example (Random Number)

Question:

Suppose X is chosen uniformly at random from the interval $[0, 10]$.

1. What is the probability density at $x = 4$?
2. What is the probability that $X \leq 4$?
3. What are the expected value and variance of X ?

Solution:

- PDF (density at $x = 4$):

$$f(4) = \frac{1}{10-0} = \frac{1}{10}$$

- CDF (up to 4):

$$P(X \leq 4) = \frac{4-0}{10-0} = \frac{4}{10} = 0.4$$

- Mean:

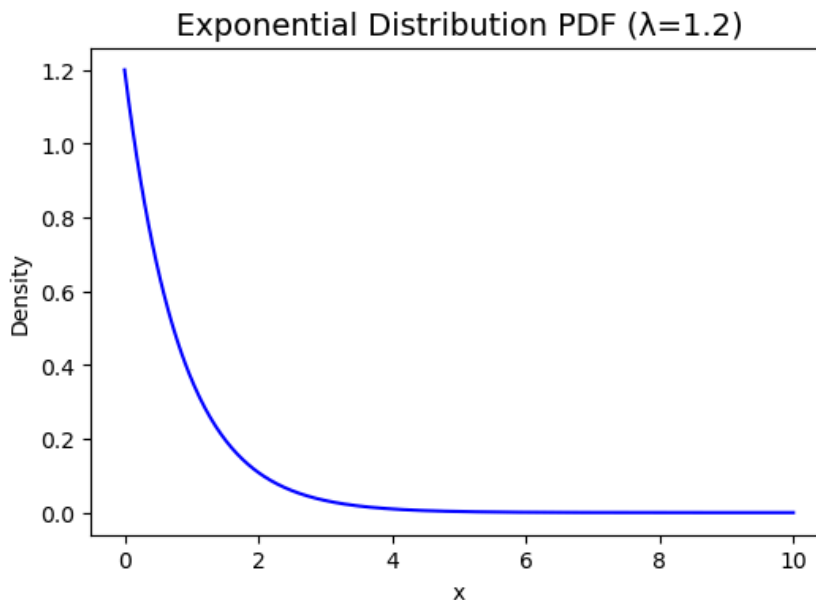
$$\mathbb{E}[X] = \frac{0+10}{2} = 5$$

- Variance:

$$\text{Var}(X) = \frac{(10-0)^2}{12} = \frac{100}{12} \approx 8.33$$



Exponential Distribution



Concept & Intuition:

The exponential distribution models the **time until the first event** in a **Poisson process** — a process where events occur **independently and at a constant average rate** over time. It is the **continuous analogue of the geometric distribution**.

It is often used to model **waiting times** (e.g., time until next arrival, email, failure, etc.).

Parameters:

- $\lambda > 0$: rate parameter (events per unit time)
-

PDF:

$$f(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0, \\ 0, & x < 0. \end{cases}$$

CDF:

$$F(x) = \begin{cases} 0, & x < 0, \\ 1 - e^{-\lambda x}, & x \geq 0. \end{cases}$$

Mean:

$$\mathbb{E}[X] = \frac{1}{\lambda}$$

Variance:

$$\text{Var}(X) = \frac{1}{\lambda^2}$$

Memoryless Property:

The exponential distribution is **memoryless**, just like the geometric distribution:

$$P(X > s + t \mid X > s) = P(X > t)$$

This means the probability of waiting **more time** doesn't depend on how long you've already waited.

Relationship with Geometric Distribution:

Both model "time until first success":

- Geometric: discrete time steps
- Exponential: continuous time
- Both are memoryless.

Relationship with Poisson Distribution (Poisson Process):

If events occur according to a Poisson process with rate λ , then:

- $X \sim \text{Exponential}(\lambda)$ = time **until the first event**
- $N(t) \sim \text{Poisson}(\lambda t)$ = number of events **up to time t**

These two are **tightly connected**:

$$P(X > t) = P(\text{no events in } [0, t]) = P(N(t) = 0)$$

Example – Email Arrival:

Let emails arrive at rate $\lambda = 2$ per hour (Poisson process).

Let X be the time until the **first email**. Then $X \sim \text{Exponential}(2)$.

Q: What is the probability that you wait more than $t = 0.5$ hours to get the first email?

- Using Exponential:

$$P(X > 0.5) = e^{-2 \cdot 0.5} = e^{-1} \approx 0.3679$$

- Using Poisson:

$$P(N(0.5) = 0) = \frac{(2 \cdot 0.5)^0 e^{-1}}{0!} = e^{-1} \approx 0.3679$$

✅ Same result!

Both expressions describe the same event: **no emails arrive in the first 0.5 hours**.

Question:

Suppose buses arrive at a rate of $\lambda = 0.2$ per minute (on average one bus every 5 minutes). Let X be the waiting time until the next bus.

1. What is the probability density at $x = 5$ minutes?
2. What is the probability of waiting at most 5 minutes (i.e., $P(X \leq 5)$)?
3. What are the expected value and variance of X ?

Solution:

- PDF (density at 5):

$$f(5) = 0.2e^{-0.2 \cdot 5} = 0.2e^{-1} \approx 0.0736$$

- CDF (up to 5):

$$P(X \leq 5) = 1 - e^{-0.2 \cdot 5} = 1 - e^{-1} \approx 0.632$$

- Mean:

$$\mathbb{E}[X] = \frac{1}{0.2} = 5$$

- Variance:

$$\text{Var}(X) = \frac{1}{0.2^2} = 25$$

Concept & Intuition:

A Poisson process models random events happening **continuously and independently over time**, with a **constant average rate**.

Assumptions

1. Events occur **one at a time**.
2. Events happen **independently**.
3. The probability of a single event in a small interval $[t, t + \Delta t]$ is approximately $\lambda \Delta t$.
4. The number of events in an interval of length t follows a **Poisson distribution** with parameter λt .

Notation

- λ : average number of events per unit time (**rate**)
- $N(t)$: number of arrivals by time t
- X_i : time between the $(i - 1)$ -th and i -th arrivals (inter-arrival time)

Key Distributions in a Poisson Process

Quantity	Distribution
Number of arrivals in time t	$N(t)$ $\sim \text{Poisson}(\lambda t)$
Time between two consecutive events	X_i $\sim \text{Exponential}(\lambda)$
Time to the k -th event	T_k $\sim \text{Gamma}(k, \lambda)$

Example

Let $\lambda = 5$ (emails per hour).

- Expected: 5 emails per hour.
- $N(2) \sim \text{Poisson}(10) \rightarrow$ number of emails in 2 hours.
- $X_1 \sim \text{Exponential}(5) \rightarrow$ time until first email.

Relationship Between Poisson and Exponential

The exponential distribution models the time until the **first** event in a Poisson process.

$$P(X > t) = e^{-\lambda t}$$

This is exactly the same as:

$$P(N(t) = 0)$$

 **Interpretation:** The probability that we **wait more than t units of time** for the first event is the same as the probability that **no events happen** by time t .

Memoryless Property of Exponential

The exponential distribution is **memoryless**:

$$P(X > s + t \mid X > s) = P(X > t)$$

Knowing that we've already waited s time units without an event doesn't change the probability of needing to wait more.

Probability Examples

1. What is the probability of exactly 3 emails in 2 hours?

$$P(N(2) = 3) = \frac{(10)^3 e^{-10}}{3!} = \frac{1000e^{-10}}{6}$$

2. What is the probability that you wait more than 30 minutes (0.5 hours) for the next email?

$$P(X > 0.5) = e^{-5 \cdot 0.5} = e^{-2.5} \approx 0.0821$$

Gamma Connection

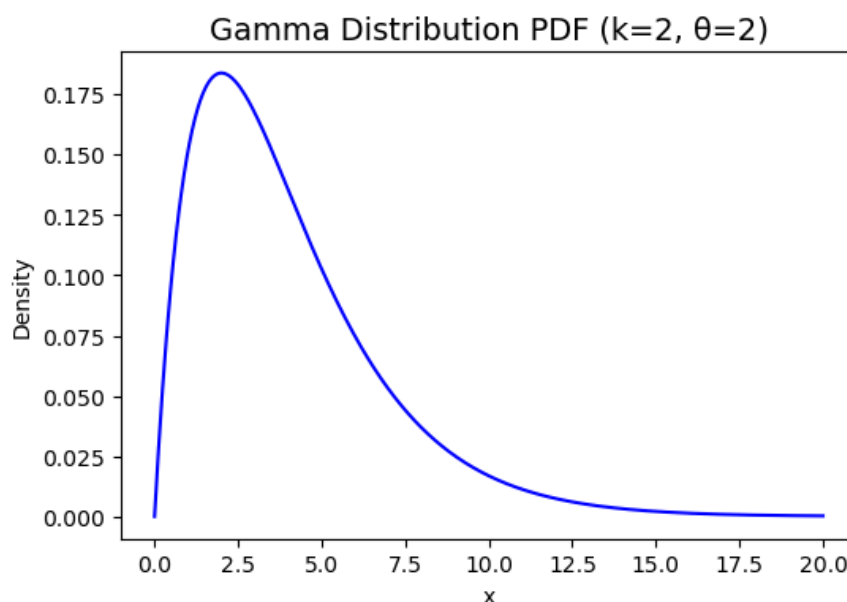
Let T_k be the time until the k -th arrival. Then:

$$T_k \sim \text{Gamma}(k, \lambda)$$

This generalizes the exponential distribution, since:

$$T_1 \sim \text{Exponential}(\lambda)$$

Gamma Distribution



Concept & Intuition:

The Gamma distribution models the **waiting time until the k -th event** in a **Poisson process** with rate λ . It is a **generalization of the Exponential distribution**, which models the time until the **first** event.

If you sum k independent exponential waiting times with the same rate λ , the total time T_k follows a Gamma distribution.

Parameters

- $k > 0$: shape parameter (can be integer or real)
- $\lambda > 0$: rate parameter (same as exponential)

If k is a positive integer, the distribution is often called the **Erlang distribution**.

PDF (Probability Density Function)

$$f(x) = \frac{\lambda^k x^{k-1} e^{-\lambda x}}{\Gamma(k)}, \quad x \geq 0$$

Here, $\Gamma(k)$ is the Gamma function.

For integers: $\Gamma(k) = (k-1)!$

CDF (Cumulative Distribution Function)

There is no simple closed-form CDF for all k , but it is given in terms of the **lower incomplete gamma function**:

$$F(x) = \frac{\gamma(k, \lambda x)}{\Gamma(k)}$$

For small integer k , it simplifies. For example, if $k = 1$ (Exponential), we get:

$$F(x) = 1 - e^{-\lambda x}$$

Special Case: Exponential Distribution

If $k = 1$, then:

$$f(x) = \lambda e^{-\lambda x}$$

Which is exactly the PDF of the **Exponential distribution**.

Mean

$$\mathbb{E}[X] = \frac{k}{\lambda}$$

Variance

$$\text{Var}(X) = \frac{k}{\lambda^2}$$

Relationship with Poisson Process

Let events arrive in a Poisson process with rate λ .

- Let T_k = time until the k -th arrival.
- Then:

$$T_k \sim \text{Gamma}(k, \lambda)$$

Duality: Gamma and Poisson

Let's look at the connection between **arrival times** and **event counts**:

- T_k : time until the k -th arrival
- $N(t)$: number of arrivals by time t

Then:

$$P(T_k > t) = P(\text{fewer than } k \text{ arrivals by time } t) = P(N(t) < k)$$

This means:

Waiting more than time t for the k -th event is the same as
having fewer than k arrivals in time t

Example

Suppose a machine breaks down after the **3rd failure**. Failures occur with rate $\lambda = 2$ per hour.

Let T_3 be the time until the 3rd failure.

- PDF:

$$f(x) = \frac{2^3 x^2 e^{-2x}}{2!} = 4x^2 e^{-2x}$$

- Mean:

$$\mathbb{E}[T_3] = \frac{3}{2} = 1.5 \text{ hours}$$

- Variance:

$$\text{Var}(T_3) = \frac{3}{2^2} = 0.75 (\text{hours})^2$$

- What is the probability that the 3rd failure happens **after 2 hours**?

Equivalent to:

$$P(T_3 > 2) = P(N(2) < 3)$$

Use Poisson distribution:

$$P(N(2) < 3) = P(N(2) = 0) + P(N(2) = 1) + P(N(2) = 2)$$

Where $N(2) \sim \text{Poisson}(2 \cdot 2) = \text{Poisson}(4)$

So:

$$P(T_3 > 2) = \sum_{k=0}^2 \frac{4^k e^{-4}}{k!}$$

Gamma Function – What Is It?

The **Gamma function**, denoted by $\Gamma(z)$, is a continuous extension of the **factorial** function to real (and even complex) numbers.

Definition

For any real number $z > 0$, the Gamma function is defined as:

$$\Gamma(z) = \int_0^{\infty} x^{z-1} e^{-x} dx$$

This is an improper integral that converges for all $z > 0$.

Connection to Factorials

For **positive integers** n , the Gamma function satisfies:

$$\Gamma(n) = (n - 1)!$$

So:

- $\Gamma(1) = 0! = 1$
 - $\Gamma(2) = 1! = 1$
 - $\Gamma(3) = 2! = 2$
 - $\Gamma(4) = 3! = 6$
 - etc.
-

Why Use the Gamma Function?

- It allows us to work with **non-integer values** of n !
 - It's needed in the formulas of **Beta** and **Gamma distributions**, where the parameters (like α , β , k) don't have to be whole numbers.
-

Examples

- $\Gamma(1/2) = \sqrt{\pi}$
This is a famous result and often used in probability.
 - $\Gamma(5.5)$ is not an integer, but it still exists and can be computed numerically.
-

Gamma Function in Distributions

1. **Gamma Distribution PDF:**

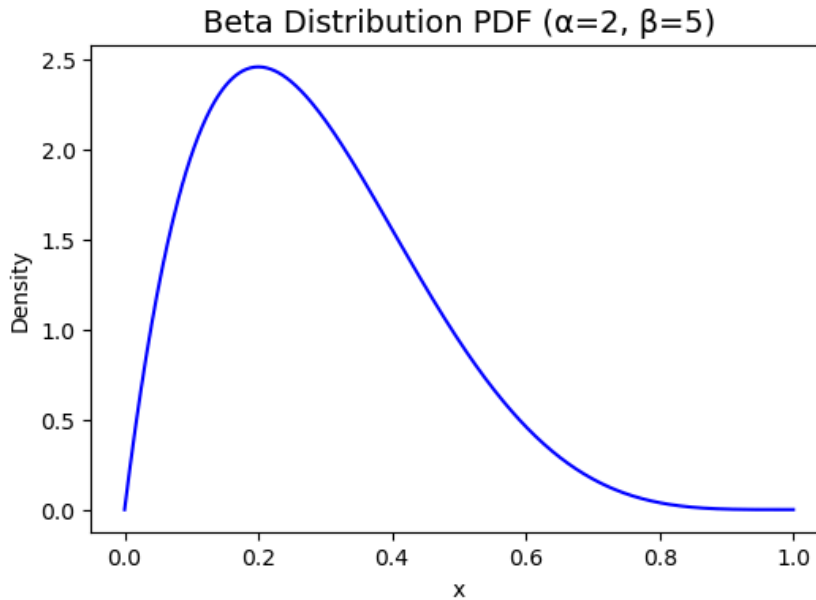
$$f(x) = \frac{\lambda^k x^{k-1} e^{-\lambda x}}{\Gamma(k)}$$

2. **Beta Distribution PDF:**

$$f(x) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1} (1 - x)^{\beta-1}$$

In both cases, $\Gamma(\cdot)$ plays a similar role to a **normalizing constant**, just like dividing by $n!$ in discrete distributions like the Binomial.

Beta Distribution



Concept & Intuition:

The Beta distribution models **random probabilities** — that is, values between 0 and 1.

It is defined on the interval $[0, 1]$ and is especially used to model **uncertainty about a probability parameter**, such as the probability p of success in a Bernoulli or Binomial process.

It is often called the "**distribution of probabilities**."

Parameters

- $\alpha > 0$: controls how peaked the distribution is near 1 (successes)
- $\beta > 0$: controls how peaked the distribution is near 0 (failures)

Together, α and β can be thought of as **prior counts** of successes and failures.

PDF (Probability Density Function)

The Beta distribution is defined as:

$$f(x) = \frac{1}{B(\alpha, \beta)} x^{\alpha-1} (1-x)^{\beta-1}, \quad 0 < x < 1$$

Where $B(\alpha, \beta)$ is the **Beta function**, defined using the **Gamma function**:

$$B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha + \beta)}$$

So:

$$f(x) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1} (1-x)^{\beta-1}$$

Mean and Variance

Mean:

$$\mathbb{E}[X] = \frac{\alpha}{\alpha + \beta}$$

Variance:

$$\text{Var}(X) = \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}$$

Relationship to the Gamma Function

If:

- $X_1 \sim \text{Gamma}(\alpha, 1)$
- $X_2 \sim \text{Gamma}(\beta, 1)$

Then:

$$\frac{X_1}{X_1 + X_2} \sim \text{Beta}(\alpha, \beta)$$

This shows the **Beta distribution as a ratio of two Gamma-distributed random variables**.

Bayesian Interpretation – Prior for Binomial

In Bayesian statistics, the Beta distribution is the **conjugate prior** for the Binomial (and Bernoulli) distribution.

Suppose we observe x successes out of n trials (Binomial likelihood), and we place a **Beta prior** on the success probability p :

- Prior: $p \sim \text{Beta}(\alpha, \beta)$
- Likelihood: $X \sim \text{Binomial}(n, p)$
- Posterior: $p \mid X \sim \text{Beta}(\alpha + x, \beta + n - x)$

So the parameters of the Beta posterior update like **adding successes and failures** to the prior.

Example

Let's say we toss a coin and observe 7 heads and 3 tails.

We place a prior: $p \sim \text{Beta}(2, 2)$ (assumes 2 prior heads and 2 prior tails)

Then:

- Posterior:

$$p \mid \text{data} \sim \text{Beta}(2 + 7, 2 + 3) = \text{Beta}(9, 5)$$

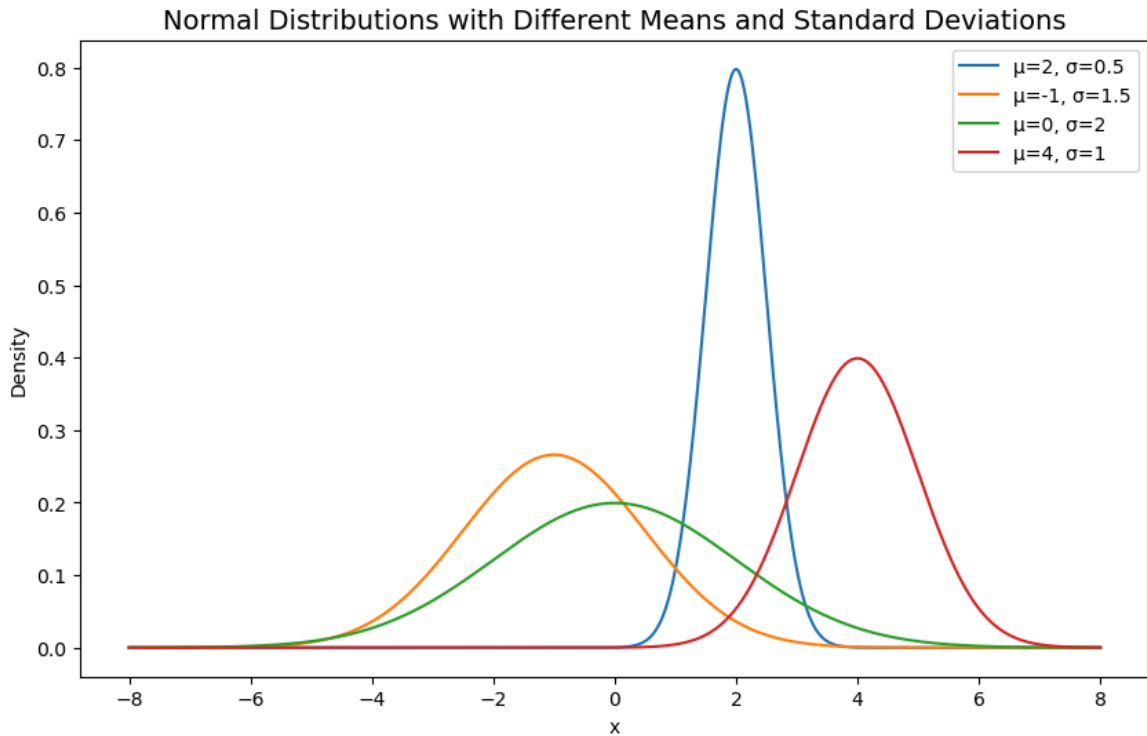
- Mean:

$$\mathbb{E}[p] = \frac{9}{9 + 5} = \frac{9}{14} \approx 0.643$$

✓ This is now your **updated belief** about the true probability of heads after observing data.

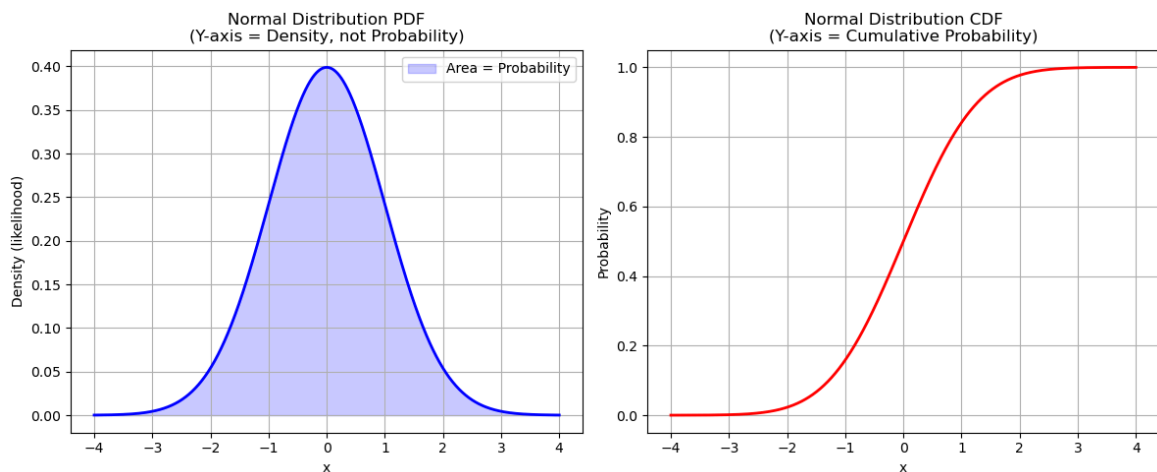
Normal (Gaussian) Distribution

Normal (Gaussian) Distribution



Concept & Intuition:

The **Normal distribution** models many naturally occurring variables (e.g., heights, test scores, errors) due to the **Central Limit Theorem**, which states that the sum (or average) of many independent random variables tends to be normally distributed — even if the original variables are not.



• Normal PDF (Probability Density Function):

- The y-axis shows **density (likelihood)**, not probability.
- This means the height of the curve at a point does **not** represent probability rather a **likelihood**.
- In fact, the PDF can take values **greater than 1** if the variance is very small.
- The **probability** is found by calculating the **area under the curve** over an interval:

$$P(a < X < b) = \int_a^b f(x) dx$$

- **Normal CDF (Cumulative Distribution Function):**

- The y-axis shows the **cumulative probability** directly.
- It always ranges from **0 to 1**.
- For a given value of x , the CDF tells you the probability that a random variable $X \leq x$:

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt$$

🔑 **Key idea:**

- In the **PDF**, probability = *area under the curve*.
- In the **CDF**, probability = *value on the y-axis*.

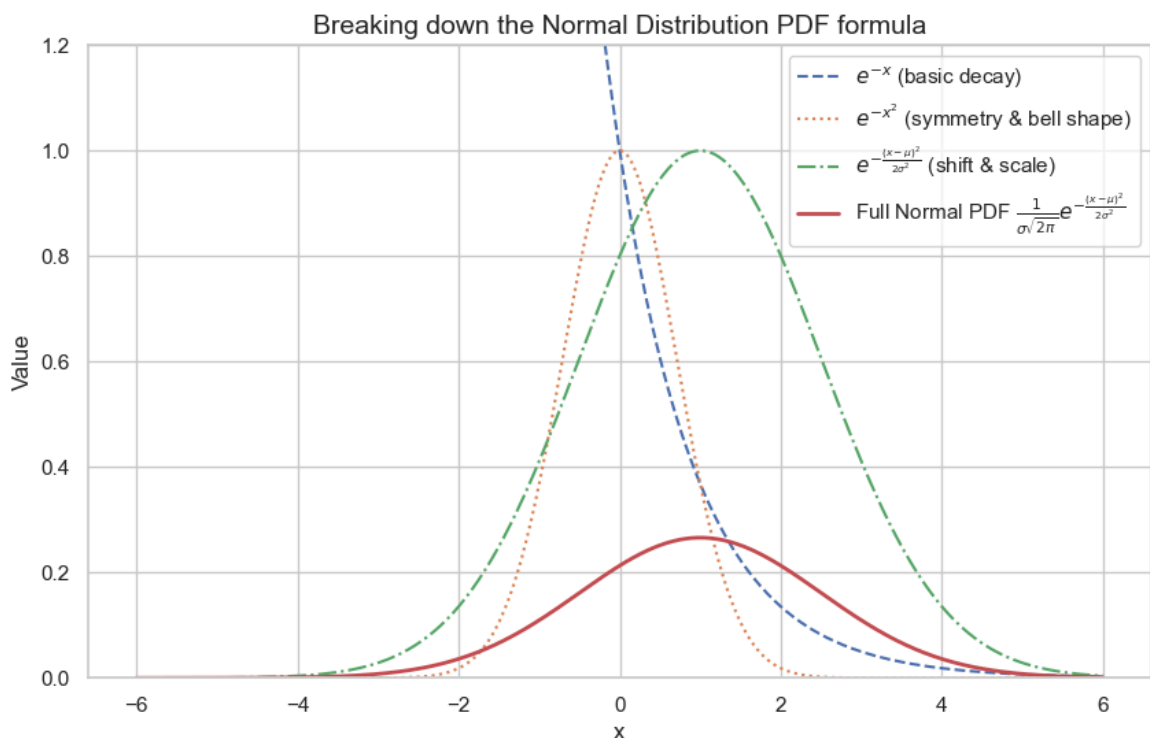
📊 PDF (Probability Density Function)

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

📐 Parameters

- μ : Mean (location)
- σ^2 : Variance (spread)
- σ : Standard deviation

Unpacking The Formula



Explanation of each part (plain text):

e^{-x} — This is a basic exponential decay function. It decreases rapidly as x increases, but it's not symmetric and only defined on positive x for a bell shape.

e^{-x^2} — By squaring x inside the exponent, we get a symmetric curve around zero because x^2 is positive whether x is positive or negative. This creates the basic bell shape but it's not normalized (area under the curve isn't 1).

$$e^{-\frac{(x-\mu)^2}{2\sigma^2}} \text{ ---}$$

The shift by μ moves the center of the bell curve to $x = \mu$.

The scaling by σ^2 controls the width (spread) of the bell:

- Larger σ means a wider, flatter curve (more spread).
- Smaller σ means a narrower, taller curve (less spread).

Normalization constant $\frac{1}{\sigma\sqrt{2\pi}}$ ---

This scales the function so the total area under the curve equals 1, making it a valid probability density function (PDF).

Without it, the curve would just show shape but not a true distribution.

Mean and Variance

- Mean:

$$\mathbb{E}[X] = \mu$$

- Variance:

$$\text{Var}(X) = \sigma^2$$

Shape and Properties

- Symmetrical, bell-shaped curve
- Defined on $(-\infty, \infty)$
- Peak at μ
- 68% of the data lies within 1 standard deviation of the mean, 95% within 2, 99.7% within 3 (**Empirical Rule**)

Use Cases

- Modeling measurement errors, natural phenomena, biological variables
- Foundation for statistical inference (e.g., confidence intervals, z-tests)

Relationship to Z-distribution

If $X \sim \mathcal{N}(\mu, \sigma^2)$, then the standardized variable:

$$Z = \frac{X - \mu}{\sigma} \sim \mathcal{N}(0, 1)$$

Z-Table (Cumulative Standard Normal Table)

Use to look up probabilities for the standard normal distribution:

 [Standard Normal Table \(Z-table\)](#)

To compute:

$$P(X \leq a) = P\left(Z \leq \frac{a - \mu}{\sigma}\right)$$

Hypothesis Testing with Normal Distribution

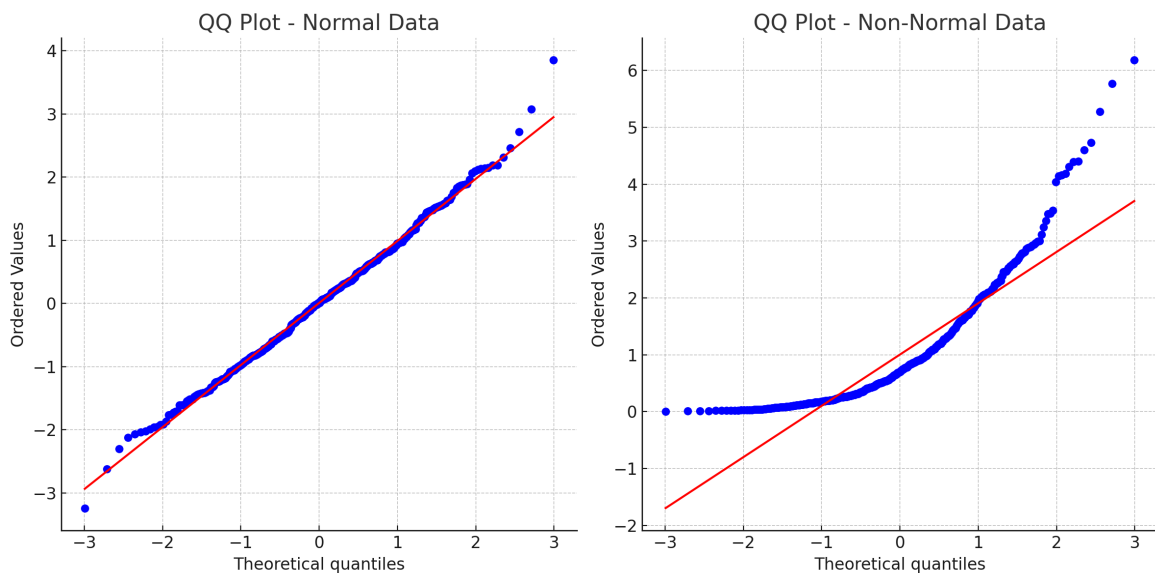
- **Z-test** for population mean when σ is known
- **Z-test** for large sample sizes (by CLT)

- **One-sample and two-sample z-tests**
- Often used to compute p-values and confidence intervals

Q-Q Plot (Quantile-Quantile Plot) - Test for Normal Distribution

- Used to assess whether your data comes from a **theoretical distribution** (usually normal).
- Compares **sample quantiles** vs. **theoretical quantiles**.
- If the data is **normally distributed**, the points in the Q-Q plot lie approximately on a **45° line**.

45° line



Mechanism:

1. Sort your sample data
2. Compute the quantiles of your sample
3. Compute the corresponding theoretical quantiles from the normal distribution
4. Plot:
 - x-axis: theoretical quantiles
 - y-axis: sample quantiles

Interpretation:

- If the points deviate systematically from the line (e.g., S-shape or curve), the data is **not normally distributed**.

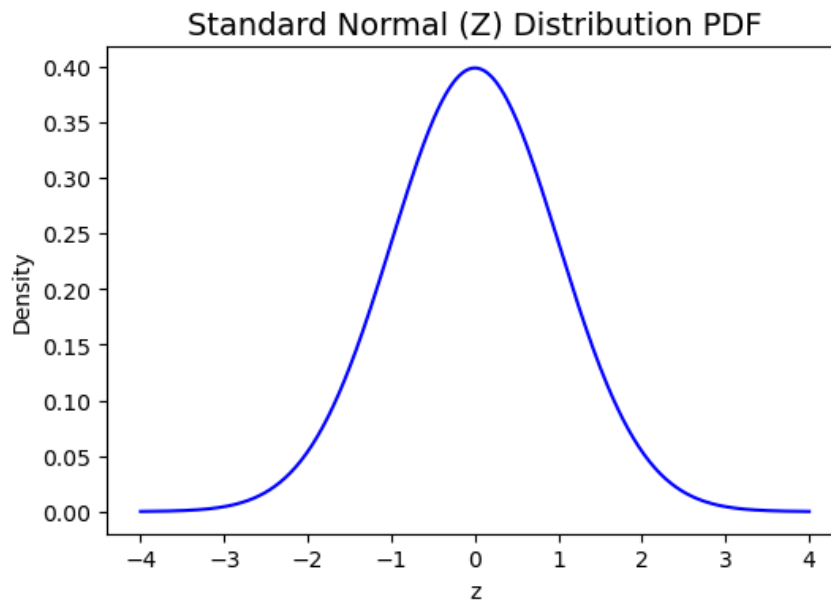
Disadvantages of Quantiles:

- Less reliable in small samples since they depend only on position, not on entire data distribution.

When to Choose What?

Situation	Preferred Measure
Symmetric Distribution	Mean
Skewed Distribution	Median, Quantiles
Categorical Data	Mode
Different Weights	Weighted Mean
Spread / Outlier Detection	Quantiles (IQR, Q1, Q3)

Standard Normal Distribution (Z-distribution)



Concept & Intuition:

The **Standard Normal distribution** is a special case of the Normal distribution with:

- Mean $\mu = 0$
- Standard deviation $\sigma = 1$

It is used to **standardize** any normal random variable using the transformation:

$$Z = \frac{X - \mu}{\sigma}$$

This lets us use **universal probability tables** (Z-tables) instead of recomputing probabilities for every normal distribution.

PDF

$$f(z) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{z^2}{2}\right)$$

Parameters

- Mean: $\mu = 0$
 - Variance: $\sigma^2 = 1$
-

Standardization

If $X \sim \mathcal{N}(\mu, \sigma^2)$, then:

$$Z = \frac{X - \mu}{\sigma} \sim \mathcal{N}(0, 1)$$

This converts any normal variable to the standard normal form.

A z-score, also known as a standard score, indicates how many standard deviations a data point is away from the mean of a distribution.

Z-Tables (Cumulative Probabilities)

Used to compute cumulative probabilities:

- Left tail:

$$P(Z \leq z)$$

- Right tail:

$$P(Z \geq z) = 1 - P(Z \leq z)$$

- Between two values:

$$P(a \leq Z \leq b) = P(Z \leq b) - P(Z \leq a)$$

 [Standard Z-Table Lookup](#)

Hypothesis Tests Using Z

- **Z-test for population mean** (when σ is known or sample size is large)
 - **Z-test for proportions**
 - **Z-confidence intervals** for means and proportions
-

Examples

Example 1: Left Tail (Less Than)

Suppose $X \sim \mathcal{N}(100, 16)$. What's the probability that $X < 106$?

Step 1: Standardize

$$Z = \frac{106 - 100}{4} = 1.5$$

Step 2: Use Z-table:

$$P(X < 106) = P(Z < 1.5) \approx 0.9332$$

 **Answer:** 93.32% of values fall below 106.

Example 2: Right Tail (Greater Than or Equal)

What's the probability that $X \geq 104$?

Step 1: Standardize

$$Z = \frac{104 - 100}{4} = 1.0$$

Step 2: Use Z-table:

$$P(X \geq 104) = P(Z \geq 1.0) = 1 - P(Z \leq 1.0) = 1 - 0.8413 = 0.1587$$

 **Answer:** 15.87% of values are greater than or equal to 104.

Example 3: Between Two Values

What's the probability that $98 \leq X \leq 106$?

Step 1: Standardize both bounds:

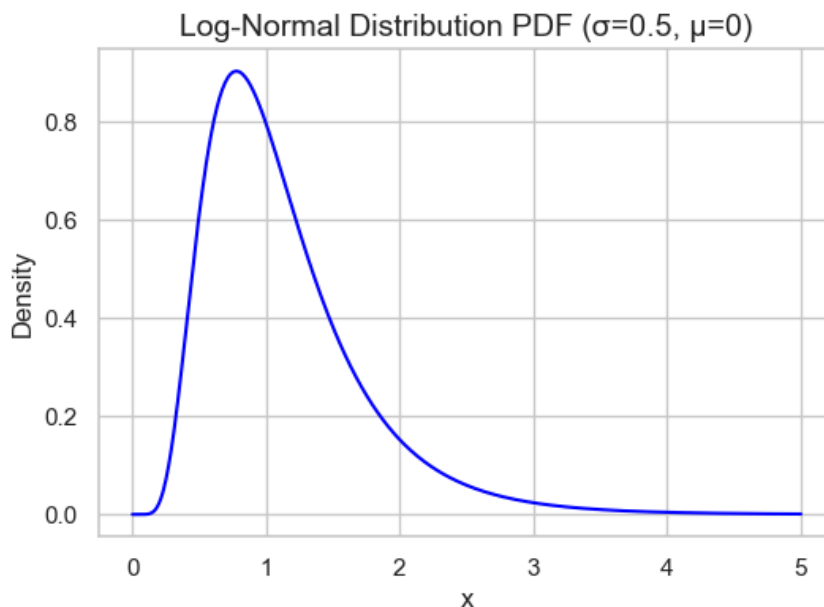
$$Z_1 = \frac{98 - 100}{4} = -0.5, \quad Z_2 = \frac{106 - 100}{4} = 1.5$$

Step 2: Use Z-table:

$$P(-0.5 \leq Z \leq 1.5) = P(Z \leq 1.5) - P(Z \leq -0.5) = 0.9332 - 0.3085 = 0.6247$$

✅ **Answer:** 62.47% of values fall between 98 and 106.

Log-Normal Distribution



Concept & Intuition:

A random variable X is **log-normally distributed** if:

$$\ln(X) \sim \mathcal{N}(\mu, \sigma^2)$$

This means that the logarithm of X follows a normal distribution.

Thus, X itself is **positively skewed**, strictly positive, and used to model variables that grow **multiplicatively**, such as:

- Stock prices
- Income
- Lengths of lifetimes (in reliability engineering)

PDF

$$f(x) = \frac{1}{x\sigma\sqrt{2\pi}} \exp\left(-\frac{(\ln x - \mu)^2}{2\sigma^2}\right), \quad x > 0$$

Parameters

- μ : the mean of $\ln(X)$
- σ^2 : the variance of $\ln(X)$

(These are not the mean and variance of X , but of $\ln X$)

Mean and Variance

Let $X \sim \text{LogNormal}(\mu, \sigma^2)$

- Mean:

$$\mathbb{E}[X] = e^{\mu + \frac{\sigma^2}{2}}$$

- Variance:

$$\text{Var}(X) = (e^{\sigma^2} - 1) e^{2\mu + \sigma^2}$$

Example

Let $X \sim \text{LogNormal}(\mu = 1, \sigma^2 = 0.25)$. Then:

- Mean:

$$\mathbb{E}[X] = e^{1+0.125} = e^{1.125} \approx 3.08$$

- Variance:

$$\text{Var}(X) = (e^{0.25} - 1) \cdot e^{2.25} \approx (1.284 - 1)(9.49) \approx 2.69$$

Use Cases

- **Stock prices** under geometric Brownian motion
 - **Biological data** (e.g., reaction times, size of organisms)
 - **Income distributions** (as an alternative to Pareto)
-

Relationship to Normal Distribution

If:

$$Y = \ln(X) \sim \mathcal{N}(\mu, \sigma^2)$$

Then:

$$X = e^Y \sim \text{LogNormal}(\mu, \sigma^2)$$

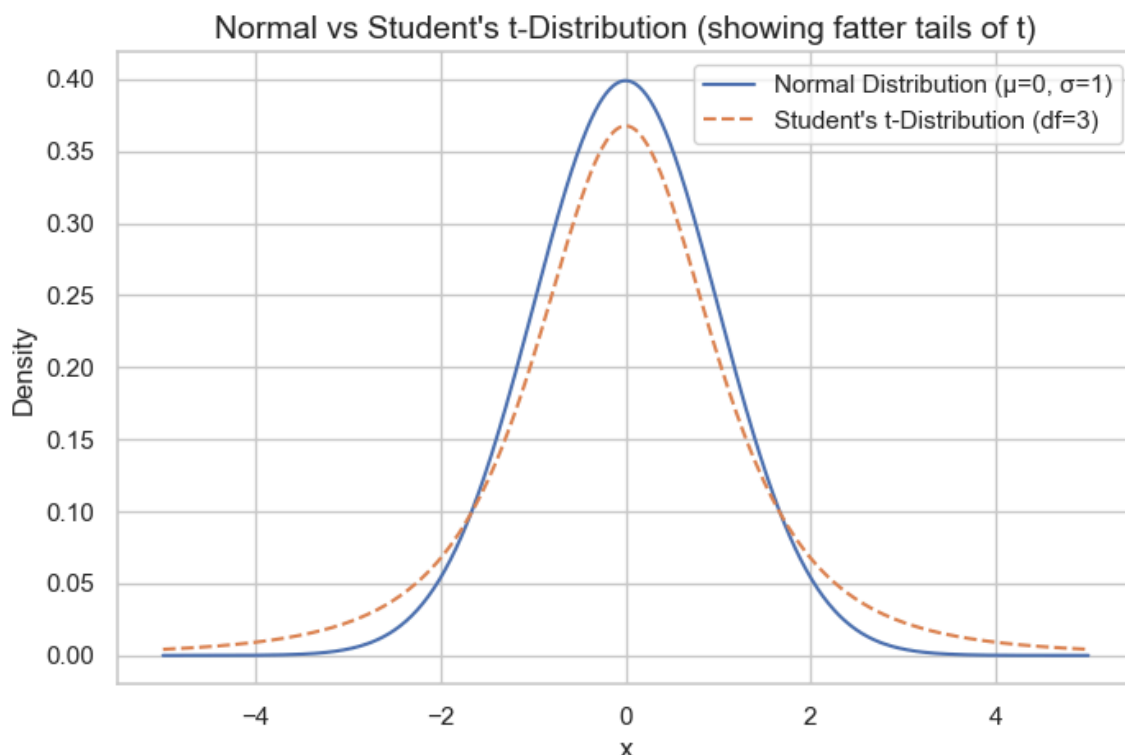
So a log-normal distribution is an **exponential transformation** of a normal distribution.

Note

- The Log-Normal is **not symmetric**.
 - It is **strictly positive**: $X > 0$
 - It has **long right tail** → good for modeling rare but large outcomes.
-

Student's t-Distribution

Student's t-Distribution



Concept & Intuition:

The t-distribution is used to estimate the population mean when the sample size is small and the population variance is **unknown**. It's wider and heavier-tailed than the standard normal distribution — to reflect extra uncertainty due to estimating σ from data.

It becomes closer to the standard normal distribution as the sample size grows.

Parameters

- $v = n - 1$: degrees of freedom (sample size minus one)
-

PDF

$$f(t) = \frac{\Gamma\left(\frac{v+1}{2}\right)}{\sqrt{v\pi} \Gamma\left(\frac{v}{2}\right)} \left(1 + \frac{t^2}{v}\right)^{-\frac{v+1}{2}}, \quad -\infty < t < \infty$$

Mean and Variance

- Mean:

$$\mathbb{E}[T] = 0 \quad (\text{for } v > 1)$$

- Variance:

$$\text{Var}(T) = \frac{v}{v-2} \quad (\text{for } v > 2)$$

Use Cases

- Small-sample confidence intervals for the mean (when σ is unknown)
- Hypothesis tests on the mean:
 - **One-sample t-test**
 - **Two-sample t-test**
 - **Paired t-test**

Example

You take a sample of $n = 10$ students' test scores and compute:

- Sample mean: $\bar{x} = 78$
- Sample standard deviation: $s = 5$

What is the 95% confidence interval for the population mean?

Step 1: Degrees of freedom:

$$v = n - 1 = 9$$

Step 2: Use t-table for $v = 9$, $t_{0.025} \approx 2.262$

Step 3: Confidence interval:

$$\bar{x} \pm t \cdot \frac{s}{\sqrt{n}} = 78 \pm 2.262 \cdot \frac{5}{\sqrt{10}} \approx 78 \pm 3.58$$

✓ So, the 95% CI is approximately: (74.42, 81.58)

t-Table (Critical Values)

 [Student's t Critical Value Table](#)

Relationship to Z

If $Z \sim \mathcal{N}(0, 1)$ and $V \sim \chi^2(v)$ are independent, then:

$$T = \frac{Z}{\sqrt{V/v}} \sim t_v$$

So the t-distribution is a **ratio of normal to chi-squared**, scaled by degrees of freedom.

Comparison: t-Distribution vs. Z-Distribution

✓ When to Use Which?

Scenario	Use Z-distribution	Use t-distribution
Known population standard deviation	✓	✗
Unknown population standard deviation	✗	✓

Scenario	Use Z-distribution	Use t-distribution
Large sample size ($n \geq 30$)	✓ (by CLT)	✓ (t becomes ~normal)
Small sample size ($n < 30$)	✗	✓

Key Formulas

- **Z-score:**

$$Z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

- **t-score:**

$$t = \frac{\bar{x} - \mu}{s / \sqrt{n}}$$

- σ : population std dev
- s : sample std dev
- n : sample size

Shape Comparison

- The **Z-distribution** is a **standard normal distribution**: perfectly symmetrical and bell-shaped.
- The **t-distribution** is also bell-shaped and symmetric but has:
 - **Heavier (fatter) tails**
 - **More variability**
 - This compensates for the uncertainty in estimating σ

Visual Shape Differences:

- For small n :
 - t-distribution is **wider, shorter peak**, and **fatter tails**
 - More probability in the extremes → **less confident**
- As $n \rightarrow \infty$:
 - t-distribution → Z-distribution

Sensitivity to Outliers

- **Z-distribution** uses fixed σ → **more sensitive to outliers**
- **t-distribution** uses sample s → **robust for small data**, compensates for variability

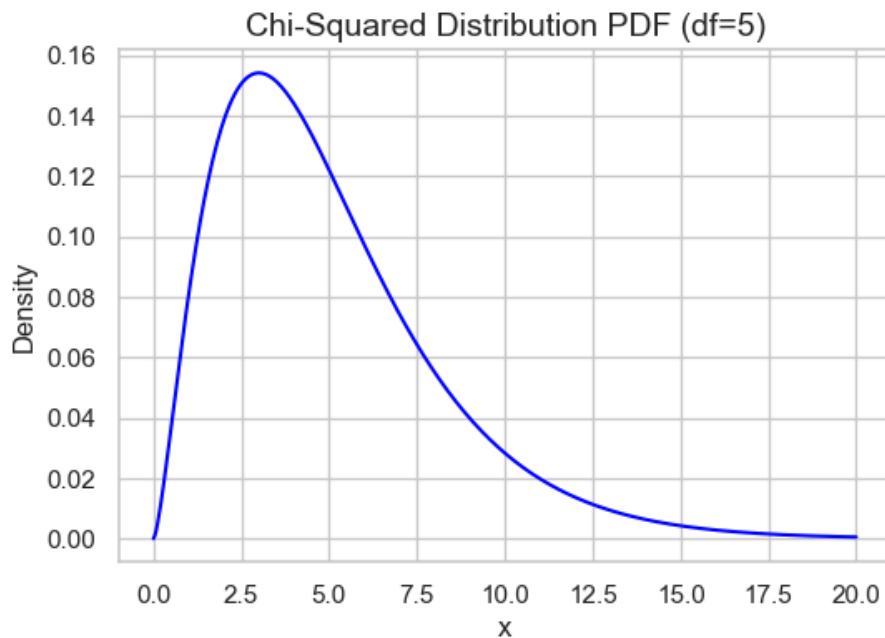
Interpretation Summary

Feature	Z-Distribution	t-Distribution
Symmetry	Yes	Yes
Tails	Thin	Thick (fat tails)
Robust to small samples	✗ No	✓ Yes
Sensitive to outliers	✓ Yes	✗ Less sensitive
Approaches Z as $n \rightarrow \infty$	—	✓ Yes

Final Rule of Thumb

If σ is unknown and your sample size is small, use the t-distribution.
Otherwise (large sample or known σ), use the Z-distribution.

Chi-Squared (χ^2) Distribution



Concept & Intuition:

The Chi-Squared distribution arises when you **sum the squares of k independent standard normal random variables**:

$$\chi_k^2 = Z_1^2 + Z_2^2 + \dots + Z_k^2, \quad Z_i \sim \mathcal{N}(0, 1)$$

It is **positively skewed** and defined only on $[0, \infty)$.

Very few real-world observations follow a chi-square distribution. The main purpose of chi-square distributions is hypothesis testing, not describing real-world distributions.

Parameters

- k : degrees of freedom (number of squared standard normal variables)

PDF

$$f(x) = \frac{1}{2^{k/2} \Gamma(k/2)} x^{(k/2)-1} e^{-x/2}, \quad x > 0$$

Mean and Variance

- Mean:

$$\mathbb{E}[X] = k$$

- Variance:

$$\text{Var}(X) = 2k$$

Use Cases

1. **Goodness-of-Fit Test** (e.g., test if data fits a distribution)
2. **Test of Independence** in contingency tables
3. **Confidence Intervals for Population Variance** (normal population)
4. **Test for Variance Equality**

Example – Goodness-of-Fit Test

Suppose you roll a die 60 times and observe the following frequencies:

Face	1	2	3	4	5	6
Obs	10	11	9	8	12	10

Expected frequency under uniform distribution = $60 / 6 = 10$

Compute:

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i} = \frac{(10 - 10)^2}{10} + \dots + \frac{(10 - 10)^2}{10} = 1.0$$

Then compare this statistic to a χ^2 distribution with $k = 6 - 1 = 5$ degrees of freedom.

Chi-Squared Table

To look up critical values:

 [Chi-Squared Table](#)

(scroll down to "Chi-Square Distribution")

Relationship to Other Distributions

- If $Z \sim \mathcal{N}(0, 1)$, then $Z^2 \sim \chi_1^2$
- If $T \sim t_v$, then:

$$T = \frac{Z}{\sqrt{\chi_v^2/v}}$$

So the t-distribution is built from both **normal** and **chi-squared** distributions.

Shape Characteristics

- As k increases, the distribution becomes more symmetric.
- Right-skewed when k is small.
- Defined only for $x > 0$
- When k is greater than two, the chi-square distribution is hump-shaped. The curve starts out low, increases, and then decreases again.



Maximum Likelihood Estimation (MLE) - Full Derivation and Explanation



Conceptual Overview

Maximum Likelihood Estimation (MLE) is a method for estimating the parameters of a statistical model. It selects the parameter values that maximize the probability (likelihood) of the observed data.

Let $x_1, x_2, \dots, x_n$ be independent and identically distributed (i.i.d.) observations from a distribution with pdf or pmf $f(x; \theta)$, where θ is the parameter (or vector of parameters) to be estimated.

Likelihood Function:

$$L(\theta) = \prod_{i=1}^n f(x_i; \theta)$$

Log-Likelihood Function:

$$\ell(\theta) = \log L(\theta) = \sum_{i=1}^n \log f(x_i; \theta)$$

MLE Estimate:

Find $\hat{\theta}$ such that: $\frac{d\ell(\theta)}{d\theta} = 0$ and solve for θ



Why Take the Log?

- Simplifies the math by converting products into sums
- Helps avoid numerical underflow for small probabilities
- Easier to differentiate when finding maxima



Univariate vs Multivariate MLE

- **Univariate:** One parameter (e.g., λ in Poisson)
- **Multivariate:** Multiple parameters (e.g., μ, σ^2 in Normal distribution)



Binomial Distribution: $X \sim \text{Bin}(n, p)$

Given fixed number of trials n , and probability of success p :

Likelihood:

$$L(p) = \prod_{i=1}^n \binom{n}{x_i} p^{x_i} (1-p)^{n-x_i}$$

Log-likelihood:

$$\ell(p) = \sum_{i=1}^n \left[\log \binom{n}{x_i} + x_i \log p + (n - x_i) \log(1 - p) \right]$$

Differentiate w.r.t. p :

$$\frac{d\ell}{dp} = \sum_{i=1}^n \left[\frac{x_i}{p} - \frac{n - x_i}{1 - p} \right]$$

Set derivative to 0:

$$\sum \left(\frac{x_i}{p} - \frac{n - x_i}{1 - p} \right) = 0$$

Solve:

$$\frac{\sum x_i}{p} = \frac{n(n - \bar{x})}{1 - p} \Rightarrow \hat{p} = \frac{\bar{x}}{n}$$

✓ Poisson Distribution: $X \sim \text{Pois}(\lambda)$

Likelihood:

$$L(\lambda) = \prod_{i=1}^n \frac{\lambda^{x_i} e^{-\lambda}}{x_i!}$$

Log-likelihood:

$$\ell(\lambda) = \sum_{i=1}^n [x_i \log \lambda - \lambda - \log(x_i!)]$$

Differentiate:

$$\frac{d\ell}{d\lambda} = \sum \left(\frac{x_i}{\lambda} - 1 \right) = \frac{1}{\lambda} \sum x_i - n$$

Set derivative to 0:

$$\frac{\sum x_i}{\lambda} = n \Rightarrow \hat{\lambda} = \bar{x}$$

✓ Normal Distribution: $X \sim \mathcal{N}(\mu, \sigma^2)$

PDF:

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

Log-likelihood:

$$\ell(\mu, \sigma^2) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum (x_i - \mu)^2$$

Differentiate w.r.t. μ :

$$\frac{\partial \ell}{\partial \mu} = \frac{1}{\sigma^2} \sum (x_i - \mu) = 0 \Rightarrow \hat{\mu} = \bar{x}$$

Differentiate w.r.t. σ^2 :

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2(\sigma^2)^2} \sum (x_i - \mu)^2 = 0$$

Solve:

$$\hat{\sigma}^2 = \frac{1}{n} \sum (x_i - \bar{x})^2$$

✓ Exponential Distribution: $X \sim \text{Exp}(\lambda)$

PDF:

$$f(x) = \lambda e^{-\lambda x}, \quad x \geq 0$$

Likelihood:

$$L(\lambda) = \prod \lambda e^{-\lambda x_i} = \lambda^n e^{-\lambda \sum x_i}$$

Log-likelihood:

$$\ell(\lambda) = n \log \lambda - \lambda \sum x_i$$

Differentiate:

$$\frac{d\ell}{d\lambda} = \frac{n}{\lambda} - \sum x_i = 0$$

Solve:

$$\hat{\lambda} = \frac{n}{\sum x_i} = \frac{1}{\bar{x}}$$

📖 Summary Table

Distribution	Parameters	MLE Estimates
Binomial	p	$\hat{p} = \frac{\bar{x}}{n}$
Poisson	λ	$\hat{\lambda} = \bar{x}$
Normal	μ, σ^2	$\hat{\mu} = \bar{x}, \hat{\sigma}^2 = \frac{1}{n} \sum (x_i - \bar{x})^2$
Exponential	λ	$\hat{\lambda} = \frac{1}{\bar{x}}$

📖 Inferential Statistics

🧠 What Is Inferential Statistics?

Inferential statistics is the branch of statistics that allows us to:

Make conclusions about a population based on data from a sample.

It goes **beyond describing data** — it helps answer questions like:

- "What is the average height of all students?"
- "Does a new drug work better than the old one?"
- "Is there a relationship between income and education?"

Inductive Reasoning Connection

Inferential statistics is based on **inductive logic**:

From specific observations (sample) → to general conclusions (population)

It involves **uncertainty** and **probabilistic thinking**, not certainty.

Example:

- If 60 out of 100 voters sampled support a candidate, we may infer ~60% of the population supports them.

Difference Between Sampling and Inference

Concept	Description
Sampling	Collecting data from a subset of the population
Inference	Drawing general conclusions from that data

What Does It Include?

- Estimating population parameters
- Constructing confidence intervals
- Performing hypothesis tests
- Predictive modeling (regression, etc.)
- Measuring uncertainty in generalizations

Hypothesis Testing – Conceptual Overview

What Is Hypothesis Testing?

Hypothesis testing is a structured statistical decision-making process to assess **claims about population parameters** using sample data.

Think of it like a **court trial**:

- The null hypothesis is the presumption of innocence.
- The data is the evidence.
- The goal is to determine whether the evidence is strong enough to reject the null.

Null vs. Alternative Hypotheses

Term	Definition
Null Hypothesis (H_0)	The status quo; assumes no effect or no difference.
Alternative Hypothesis (H_1 or H_a)	The research claim; assumes a real effect or difference.

Examples:

- Drug Test:
 $H_0: \mu_{\text{new}} = \mu_{\text{old}}$
 $H_1: \mu_{\text{new}} \neq \mu_{\text{old}}$

- Ad Campaign:

$$H_0: \mu = \mu_0$$

$$H_1: \mu > \mu_0$$

What Is a p-Value?

The **p-value** answers the question:

"Assuming H_0 is true, how likely is it to observe data as extreme or more extreme than what I observed?"

- **Low p-value ($< \alpha$):** Evidence **against** $H_0 \rightarrow$ Reject H_0
- **High p-value ($\geq \alpha$):** Evidence **is not strong enough** $\rightarrow$ Fail to reject H_0

! We never accept H_1 — we only "fail to reject" or "reject" H_0 .

Common Significance Levels (α)

α	Interpretation
0.05	5% risk of Type I error (standard)
0.01	More conservative
0.10	More exploratory or lenient

Decision rule:

- If $p < \alpha \rightarrow$ Reject H_0
- If $p \geq \alpha \rightarrow$ Fail to reject H_0

Understanding the p-value Threshold ($\alpha = 0.05$)

The **p-value** measures the probability of observing results as extreme as the ones obtained, **assuming the null hypothesis is true**.

A common significance level is:

$$\alpha = 0.05$$

This means we are willing to accept a **5% chance of making a Type I error**, i.e., **rejecting the null hypothesis when it is actually true**.

Intuitive Interpretation:

- If we **repeat the experiment 100 times**, and the null hypothesis is actually true:
 - About **95 times**, the results **will not be extreme enough** to reject the null.
 - About **5 times**, just by **random chance**, we might see results so extreme that **we wrongly reject the null**.

When $p < 0.05$:

- The observed result is considered "**statistically significant**".
- It means:

"If the null hypothesis were true, the chance of observing this result (or more extreme) is less than 5%."

This makes us **doubt the null hypothesis** and consider that there might be a real effect.

⚠ Note:

- $p < 0.05$ does not prove your hypothesis is true.
- It only indicates that your result is **unlikely due to chance** under the null hypothesis.

✗ Type I and Type II Errors

Decision \ Truth	H_0 is True	H_0 is False
Reject H_0	✗ Type I Error (false positive)	✓ Correct (true effect)
Fail to Reject H_0	✓ Correct (true negative)	✗ Type II Error (false negative)

- **Type I Error (α):** False alarm — detecting an effect that doesn't exist.
- **Type II Error (β):** Miss — failing to detect a real effect.

📦 What Is a Confidence Interval (CI)?

A **confidence interval** gives a plausible range of values for a population parameter.

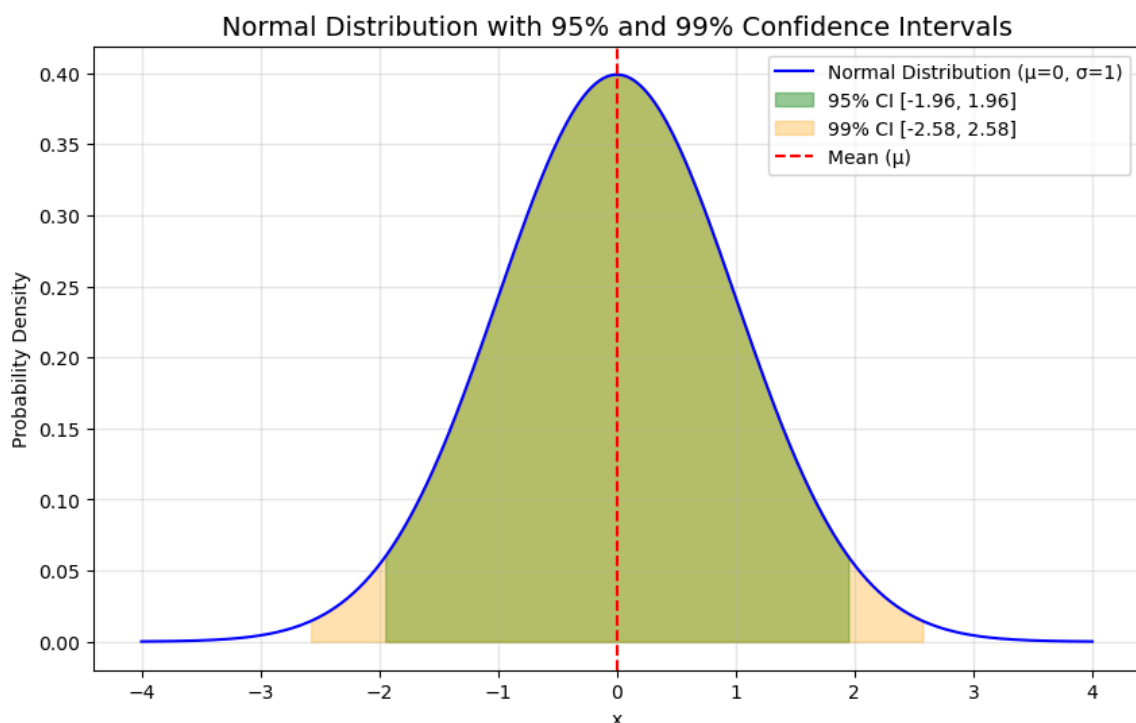
A 95% CI means:

"If we repeated this experiment many times, 95% of the CIs would contain the true population value."

CI = Estimate $\pm$ Margin of Error

📊 Confidence Intervals (CI): 95% and 99%

A **confidence interval (CI)** gives a range of values that is likely to contain the **true population parameter** (e.g., mean) with a certain level of confidence (like 95% or 99%).



✓ Formula for CI of the Mean (with known or large sample size)

Assuming:

- $\bar{x}$ is the sample mean
- z^* is the z-critical value (depends on confidence level)
- σ is the population standard deviation (or sample standard deviation s if n is large)
- n is the sample size

$$CI = \bar{x} \pm z^* \cdot \frac{\sigma}{\sqrt{n}}$$

Common z -values:

Confidence Level	z^*
95%	1.96
99%	2.576

Example:

If:

- $\bar{x} = 50$
- $\sigma = 10$
- $n = 100$

Then the **95% CI** is:

$$50 \pm 1.96 \cdot \frac{10}{\sqrt{100}} = 50 \pm 1.96 \Rightarrow [48.04, 51.96]$$

And the **99% CI** is:

$$50 \pm 2.576 \cdot \frac{10}{\sqrt{100}} = 50 \pm 2.576 \Rightarrow [47.42, 52.58]$$

Note:

- If the **sample size is small** and σ is unknown $\rightarrow$ use the **t -distribution** instead of z .
- Wider confidence intervals = **more certainty**, but **less precision**.

Relationship Between CI and p-Value

- If the CI **excludes** the null value $\rightarrow p < \alpha \rightarrow$ Significant
- If the CI **includes** the null value $\rightarrow p \geq \alpha \rightarrow$ Not significant

Common Confidence Levels

Confidence Level	Typical Use Case
95%	Standard reporting level
99%	High certainty required
90%	Exploratory or early research

Comparing Undirected vs Directed Hypotheses and Two-tailed vs One-tailed t-tests

1. Hypothesis Types

Type	Definition	Form	Example Research Question
Undirected (Non-directional)	Predicts a difference or relationship, but does not specify <i>direction</i> .	$H_0 : \mu_1 = \mu_2$ $H_a : \mu_1 \neq \mu_2$	"Does caffeine affect reaction time?" (Could be faster or slower)
Directed (Directional)	Predicts a difference in a <i>specific direction</i> .	$H_0 : \mu_1 \leq \mu_2$ $H_a : \mu_1 > \mu_2$ (or the reverse)	"Does caffeine decrease reaction time?" (Only looking for improvement/faster times)

Key Point:

- **Undirected** → both extremes of the distribution matter.
- **Directed** → only one extreme matters.

2. t-Tests and Rejection Regions at $\alpha = 0.05$

Two-tailed test (for undirected hypotheses)

- $\alpha = 0.05$ split equally into two tails $\Rightarrow 0.025$ in each tail.
- **Critical region:**
 - Left tail: $t < -t_{\alpha/2}$
 - Right tail: $t > +t_{\alpha/2}$
- Reject H_0 if the test statistic falls into *either* tail.

Visual:

|----2.5%----| 95% |----2.5%----|

<---Reject---Keep H_0 ---Keep H_0 ---Reject--->

One-tailed test (for directed hypotheses)

- $\alpha = 0.05$ entirely in **one tail**.
- **Critical region:**
 - Upper-tail: $t > t_{\alpha}$ (if predicting $\mu_1 > \mu_2$)
 - Lower-tail: $t < -t_{\alpha}$ (if predicting $\mu_1 < \mu_2$)
- Reject H_0 only if the test statistic falls into the single predicted tail.

Visual (upper-tail example):

| 95% |----5%----| Keep H_0 -----Reject

3. Real-World Example: Caffeine & Reaction Time

Scenario:

You test $n_1 = 20$ people without caffeine and $n_2 = 20$ with caffeine.

Case A — Undirected (two-tailed)

- $H_0: \mu_{\text{caffeine}} = \mu_{\text{no caffeine}}$
- $H_a: \mu_{\text{caffeine}} \neq \mu_{\text{no caffeine}}$
- Degrees of freedom: $df \approx 38$
- t_{critical} at $\alpha = 0.05$ (two-tailed) $\approx \pm 2.024$
- Reject H_0 if $t < -2.024$ or $t > 2.024$.

Case B — Directed (one-tailed)

- $H_0: \mu_{\text{caffeine}} \geq \mu_{\text{no caffeine}}$
- $H_a: \mu_{\text{caffeine}} < \mu_{\text{no caffeine}}$ (lower-tail test)
- Degrees of freedom: $df \approx 38$
- t_{critical} at $\alpha = 0.05$ (one-tailed) ≈ -1.685
- Reject H_0 if $t < -1.685$.

4. Visual Summary

Hypothesis Type	t-Test	α Split	Critical Region(s)	Portion of Distribution Rejected
Undirected	Two-tailed	0.025 each tail	$t < -t_{\alpha/2}$ or $t > t_{\alpha/2}$	Both extremes (2.5% left + 2.5% right)
Directed	One-tailed	0.05 in one tail	$t < -t_{\alpha}$ or $t > t_{\alpha}$	Only one extreme (5%)



General Steps of a Statistical Test



Purpose

The purpose of a statistical test is to determine whether sample data provide enough evidence to reject a null hypothesis about a population.



Steps of Hypothesis Testing

1. State the Hypotheses

- Null hypothesis (H_0): no effect or difference
- Alternative hypothesis (H_1 or H_a): there is an effect or difference

2. Choose a Significance Level (α)

- Typical choices: 0.05, 0.01, 0.10

3. Choose the Right Test

- Based on data type, sample size, known variance
- Examples: Z-test, t-test, chi-squared test

4. Compute the Test Statistic

- Standardizes the distance between sample and hypothesized value

Examples:

- Z-test:

$$Z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$

- t-test:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

5. Find the p-Value

- The probability of observing this result or more extreme, assuming H_0 is true

6. Make a Decision

- If $p < \alpha$: Reject H_0 (significant)
- If $p \geq \alpha$: Fail to reject H_0 (not significant)

7. Interpret in Context

- What does this decision mean in the real-world scenario?

Key Quantities You Compute

Term	Description
Test Statistic	Measures how far your result is from H_0 (in standard units)
p-Value	Probability of getting a result at least this extreme under H_0
Critical Value	Optional: cutoff value based on α for decision-making
Confidence Interval	Optional: interval estimate for the population parameter

 All of these steps are guided by **inferential statistics** — using your sample to draw conclusions about a broader population.

t-Tests: One-sample, Paired, and Independent Samples

What is a t-test?

A **t-test** is a statistical hypothesis test used to determine whether a sample mean is significantly different from:

- A population mean (one-sample)
- Another sample mean (independent two-sample)
- A paired observation (paired sample)

We use the **t-distribution** instead of the normal distribution when:

- The population standard deviation is **unknown**
- The sample size is **small** (typically $n < 30$)

General Steps for Any t-Test

1. State the Hypotheses

- Null hypothesis (H_0): Assumes no difference
- Alternative hypothesis (H_1): Assumes a difference

2. Choose Significance Level

- Common values: $\alpha = 0.05, 0.01$

3. Calculate the Test Statistic (t)

4. Find the Degrees of Freedom (df)

5. Compare to Critical Value OR Find p-Value

- Use a [t-table](#)
- If $|t| > t_{\text{critical}}$: reject H_0
- Or: if $p < \alpha$: reject H_0

1 One-Sample t-Test

When to Use:

- You want to test whether the mean of a sample differs from a known or hypothesized population mean.

Example:

A machine is supposed to fill bottles with 500 ml of water. You take a sample of 10 bottles and test if the average fill is different.

Test Statistic:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

- $\bar{x}$: sample mean
- μ_0 : hypothesized population mean
- s : sample standard deviation
- n : sample size
- $df = n - 1$

2 Paired Samples t-Test

When to Use:

- You have **two related samples** (e.g., before-and-after for same people) and want to test if the **mean difference** is significant.

Example:

Test scores before and after a training program.

Test Statistic:

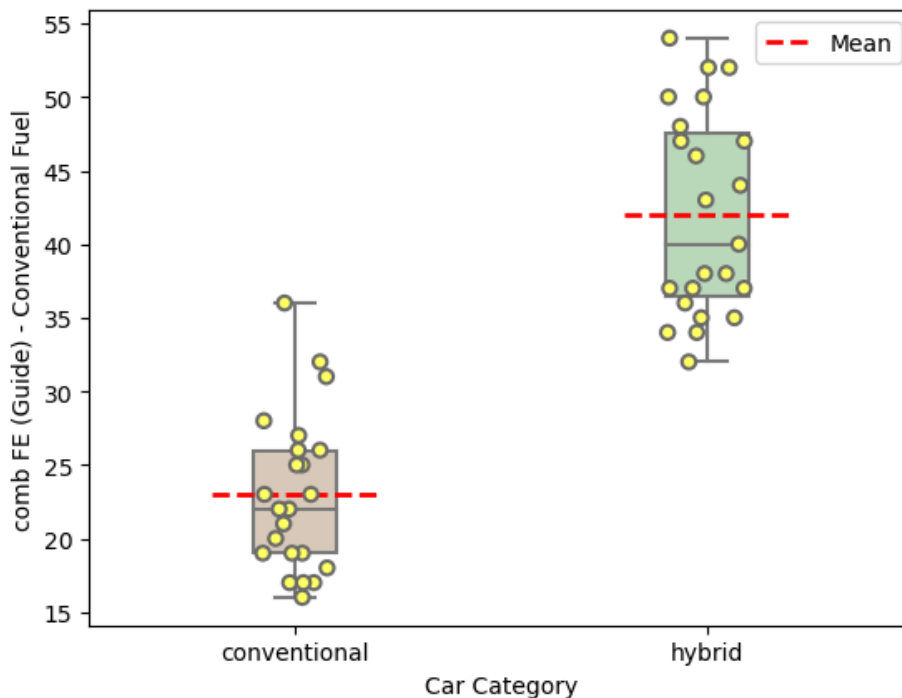
$$t = \frac{\bar{d}}{s_d/\sqrt{n}}$$

- $\bar{d}$: mean of the differences
- s_d : standard deviation of the differences
- n : number of pairs
- $df = n - 1$

3 Two Independent Samples t-Test

When to Use:

- You have **two independent groups** and want to compare their means.



Example:

Do men and women differ in average income?

Test Statistic:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s_p \cdot \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

Where pooled standard deviation is:

$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}$$

- s_1, s_2 : sample standard deviations
- n_1, n_2 : sample sizes
- $df = n_1 + n_2 - 2$

How to Get the p-Value or Critical Value

After computing t , you can either:

Option 1: Compare to Critical Value

Use the t-distribution table with your degrees of freedom and significance level α to find the **critical value**:

- Two-tailed test: compare $|t|$ to $t_{\alpha/2, df}$
- One-tailed test: compare t to $t_{\alpha, df}$

If:

- $|t| > t_{\text{critical}}$: Reject H_0
- Otherwise: Fail to reject H_0

✔ Option 2: Find the p-value

- Use a **t-table** or calculator to find the area under the curve beyond your computed t .
- If $p < \alpha$: Reject H_0
- If $p \geq \alpha$: Fail to reject H_0

Summary

Test Type	When to Use	Degrees of Freedom	Hypotheses Format
One-Sample t-Test	Compare sample to known population	$n - 1$	$H_0 : \mu = \mu_0$
Paired t-Test	Compare two related measurements	$n - 1$ (pairs)	$H_0 : \mu_d = 0$
Two-Sample t-Test	Compare two independent sample means	$n_1 + n_2 - 2$	$H_0 : \mu_1 = \mu_2$

 Always state results clearly:

"There is (or is not) enough evidence to suggest a difference between the groups..."

Two-Sample Independent t-Test – Numerical Example

Scenario

A researcher compares daily screen time (in hours) between 8 men and 8 women:

- **Men:** 6, 5, 7, 5, 6, 6, 7, 6
- **Women:** 8, 9, 7, 8, 10, 9, 7, 9

1. Hypotheses and Assumptions

- $H_0: \mu_{\text{men}} = \mu_{\text{women}}$
- $H_1: \mu_{\text{men}} \neq \mu_{\text{women}}$ (two-tailed)

1. Independence

- Observations are independent (different subjects in independent designs; repeated measures are independent across participants).

2. Normality

- The dependent variable is approximately normally distributed within each group/condition.

3. Homogeneity of variances

- For independent designs: variance is similar across groups.
- For repeated measures: the differences between conditions have equal variance (**sphericity**).

2. Sample Statistics

- $\bar{x}_1 = 6.0, s_1 \approx 0.756, n_1 = 8$
- $\bar{x}_2 = 8.375, s_2 \approx 0.916, n_2 = 8$

3. Pooled Standard Deviation

$$s_p = \sqrt{\frac{(7)(0.756)^2 + (7)(0.916)^2}{14}} \approx 0.837$$

4. Test Statistic

$$t = \frac{6.0 - 8.375}{0.837 \cdot \sqrt{\frac{1}{8} + \frac{1}{8}}} \approx -5.67$$

5. Critical Value and Decision

- Degrees of freedom: $df = 14$
 - $t_{\text{critical}} = 2.145$ (two-tailed, $\alpha = 0.05$)
 - Since $|-5.67| > 2.145$, we reject H_0
-



Conclusion

There is **strong evidence** that men and women differ in mean screen time.
Women appear to have significantly higher screen time than men.



Z-Test for a Population Mean



Concept:

A **Z-test** compares a sample mean to a known population mean when the **population standard deviation (σ) is known**, or when the sample size is large enough (usually $n \geq 30$) for the sample standard deviation to approximate σ .



Steps:

1. State Hypotheses:

- $H_0 : \mu = \mu_0$
- $H_1 : \mu \neq \mu_0$ (two-tailed) or one-sided

2. Calculate Test Statistic:

$$Z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$$

- $\bar{x}$: sample mean
- μ_0 : hypothesized population mean
- σ : population standard deviation (known)
- n : sample size

3. Find the p-value:

- Use the standard normal distribution (Z-distribution) tables or software.

4. Make Decision:

- If $p < \alpha$: reject H_0
- Else: fail to reject H_0

✓ Example:

A machine fills bottles to 500 ml exactly. The population SD is known to be 5 ml. A sample of 40 bottles has an average fill of 496 ml. Test at $\alpha = 0.05$ if the machine is underfilling.

Calculation:

$$Z = \frac{496 - 500}{5/\sqrt{40}} = \frac{-4}{0.79} \approx -5.06$$

- Critical Z-value for two-tailed $\alpha = 0.05$ is ± 1.96 .

Since $|Z| = 5.06 > 1.96$, we reject H_0 .

Conclusion:

There is strong evidence that the machine is underfilling bottles.

Chi-Squared Goodness-of-Fit Test

Concept:

A **Chi-Squared test** compares observed frequencies with expected frequencies for categorical data.

Used when:

- You want to test if a single categorical variable matches a theoretical distribution.

Types of Chi-Square Tests

Type	Purpose	Example
Goodness-of-Fit Test	Compare observed vs. expected frequencies	Is a die fair?
Test of Independence	Test if two categorical variables are related	Is gender related to voting behavior?

Example Scenario:

Test whether a 6-sided die is fair.

You roll it 60 times and get:

Face	Observed Frequency	Expected Frequency
1	5	10
2	8	10
3	9	10

Face	Observed Frequency	Expected Frequency
4	12	10
5	10	10
6	16	10

Step-by-Step:

1. Hypotheses

- H_0 : The die is fair ($P_1 = P_2 = \dots = P_6$)
- H_1 : The die is not fair (at least one P_i is different)

2. Test Statistic:

$$\chi^2 = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i}$$

Where:

- O_i : Observed frequency
- E_i : Expected frequency

3. Calculation:

$$\chi^2 = \frac{(5-10)^2}{10} + \frac{(8-10)^2}{10} + \frac{(9-10)^2}{10} + \frac{(12-10)^2}{10} + \frac{(10-10)^2}{10} + \frac{(16-10)^2}{10}$$

$$\chi^2 = \frac{25 + 4 + 1 + 4 + 0 + 36}{10} = \frac{70}{10} = 7.0$$

4. Degrees of Freedom

$$df = k - 1 = 6 - 1 = 5$$

5. Compare to Critical Value

- At $\alpha = 0.05$, $df = 5 \rightarrow$ critical value ≈ 11.07
- Since $7.0 < 11.07$, **fail to reject** H_0

Conclusion:

There is **no strong evidence** that the die is unfair.

Bonus: Chi-Square Test of Independence

If you have a **contingency table** (e.g., gender vs. vote preference), you can also use a Chi-Squared test to assess whether **two categorical variables are independent**.

In that case:

- H_0 : The two variables are **independent**
- H_1 : The two variables are **associated**

You compute expected counts with:

$$E_{ij} = \frac{(\text{Row Total}_i)(\text{Column Total}_j)}{\text{Grand Total}}$$

Then use the same χ^2 formula:

$$\chi^2 = \sum_i \sum_j \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

This is useful in survey data, marketing analysis, political studies, and more!

Analysis of Variance

ANOVA (Analysis of Variance) is a statistical method used to compare means between three or more groups to see if at least one group mean is different. It works by comparing variation between groups to variation within groups — if the between-group variation is large relative to the within-group variation, we have evidence that not all group means are the same.

You can think of one-way ANOVA as an extension of the independent samples t-test:

The independent t-test compares the means of two groups.

ANOVA generalizes this idea so you can compare three or more groups in a single test.

Instead of running multiple t-tests (which increases the risk of false positives), ANOVA lets you test all groups at once while keeping the overall error rate at the desired significance level.

ANOVA: One-way vs Two-way, With and Without Repeated Measures

1. Overview

We can distinguish ANOVA types by:

1. **Number of factors** (independent variables): One-way vs Two-way
2. **Design**: Independent measures vs Repeated measures

Case	Name	# Factors	Repeated Measures?	Example Question
1	One-way ANOVA (Independent)	1	No	Does test score differ between three teaching methods, each taught to a different group?
2	One-way ANOVA (Repeated Measures)	1	Yes	Does heart rate change for the same people after 0 mg, 100 mg, 200 mg caffeine?
3	Two-way ANOVA (Independent)	2	No	Do exam scores differ by teaching method and test difficulty , with different students in each combination?
4	Two-way ANOVA (Repeated Measures)	2	Yes	Does reaction time differ by caffeine level and time of day in the same participants?

2. Hypotheses

Case 1 — One-way ANOVA (Independent)

- **Null hypothesis (H_0):** All group means are equal.
 $\mu_1 = \mu_2 = \mu_3 \dots$
 - **Alternative hypothesis (H_a):** At least one group mean is different.
-

Case 2 — One-way ANOVA (Repeated Measures)

- H_0 : The mean response is the same under all conditions.
 - H_a : At least one condition produces a different mean response.
-

Case 3 — Two-way ANOVA (Independent)

We test **three hypotheses**:

1. Main effect of Factor A:

- H_0 : No difference in means across levels of Factor A.
- H_a : At least one level of Factor A has a different mean.

2. Main effect of Factor B:

- H_0 : No difference in means across levels of Factor B.
- H_a : At least one level of Factor B has a different mean.

3. Interaction effect ($A \times B$):

- H_0 : The effect of Factor A does not depend on Factor B.
 - H_a : The effect of Factor A depends on Factor B.
-

Case 4 — Two-way ANOVA (Repeated Measures)

- Same hypotheses as Case 3, but at least one factor is measured on the same subjects repeatedly.
-

3. Assumptions (All Cases)

1. Independence

- Observations are independent (different subjects in independent designs; repeated measures are independent across participants).

2. Normality

- The dependent variable is approximately normally distributed within each group/condition.

3. Homogeneity of variances

- For independent designs: variance is similar across groups.
 - For repeated measures: the differences between conditions have equal variance (**sphericity**).
-

4. Data Types

Factors (Independent Variables)

- **Type:** Categorical (nominal or ordinal with few levels)
- **Examples:**
 - Teaching method: Lecture, Online, Hybrid (nominal)
 - Caffeine dose: 0 mg, 100 mg, 200 mg (ordinal)
 - Time of day: Morning, Afternoon, Evening (nominal)

Dependent Variable (Parameter being tested)

- **Type:** Continuous numerical (interval or ratio)

- **Examples:**
 - Exam score (%)
 - Reaction time (ms)
 - Heart rate (BPM)
 - Weight (kg)

5. Real-World Scenarios

Case 1 — One-way ANOVA (Independent)

A school tests whether **math scores** differ among students taught using three **different teaching methods**. Each method is assigned to a different group.

Case 2 — One-way ANOVA (Repeated Measures)

A sports scientist measures **VO₂ max** for the **same athletes** under three training programs: no extra training, strength training, and endurance training.

Case 3 — Two-way ANOVA (Independent)

A company tests whether **productivity** differs by **office layout** (open, cubicle, remote) and **lighting condition** (bright vs. dim), with different employees in each combination.

Case 4 — Two-way ANOVA (Repeated Measures)

A psychologist tests whether **reaction time** is affected by **caffeine level** (0 mg, 100 mg, 200 mg) and **time of day** (morning vs. evening) in the **same participants**, measuring all combinations.

One-way ANOVA — steps & formulas

1. Notation

Let there be k groups, group i has n_i observations, and total $N = \sum_{i=1}^k n_i$. Individual observations are x_{ij} .

2. Grand mean

$$G = \frac{\sum_{i=1}^k \sum_{j=1}^{n_i} x_{ij}}{N}$$

3. Group means

For each group i :

$$M_i = \frac{\sum_{j=1}^{n_i} x_{ij}}{n_i}$$

4. Sum of squares between groups

$$SS_{btw} = \sum_{i=1}^k n_i (M_i - G)^2$$

5. Sum of squares within groups

$$SS_{wi} = \sum_{i=1}^k \sum_{j=1}^{n_i} (x_{ij} - M_i)^2$$

6. (Optional) Total sum of squares (check)

$$SS_{tot} = \sum_{i=1}^k \sum_{j=1}^{n_i} (x_{ij} - G)^2 \quad \text{and} \quad SS_{tot} = SS_{btw} + SS_{wi}$$

7. Degrees of freedom

$$df_{btw} = k - 1, \quad df_{wi} = N - k, \quad df_{tot} = N - 1$$

8. Mean squares

$$MS_{btw} = \frac{SS_{btw}}{df_{btw}} \quad \text{and} \quad MS_{wi} = \frac{SS_{wi}}{df_{wi}}$$

9. F-statistic (one-way ANOVA)

$$F = \frac{MS_{btw}}{MS_{wi}}$$

10. Example (numbers from your screenshot)

$$k = 3, \quad n_1 = n_2 = n_3 = 8, \quad N = 24$$

Group means and grand mean:

$$M_1 = 2.5, \quad M_2 = 8, \quad M_3 = 5.5, \quad G = 5.3$$

Sum of squares (as shown in screenshot):

$$SS_{btw} = 8((2.5 - 5.3)^2 + (8 - 5.3)^2 + (5.5 - 5.3)^2) = 121.33$$

$$SS_{wi} = 36$$

Degrees of freedom:

$$df_{btw} = 3 - 1 = 2, \quad df_{wi} = 24 - 3 = 21$$

Mean squares:

$$MS_{btw} = \frac{121.33}{2} = 60.67, \quad MS_{wi} = \frac{36}{21} = 1.71$$

F value:

$$F = \frac{60.67}{1.71} = 35.39$$

Two-way ANOVA — formulas and worked example

Notation (general)

- Let factor A have p levels, factor B have q levels, and there be n replicates per cell (balanced design).
- Observations: x_{mij} where $i = 1, \dots, p$ (A levels), $j = 1, \dots, q$ (B levels), $m = 1, \dots, n$ (replicates).
- Cell mean (A level i , B level j):

$$\overline{AB_{ij}} = \frac{1}{n} \sum_{m=1}^n x_{mij}$$

- Marginal means (factor A and factor B):

$$\overline{A_{i.}} = \frac{1}{q} \sum_{j=1}^q \overline{AB_{ij}}, \quad \overline{B_{.j}} = \frac{1}{p} \sum_{i=1}^p \overline{AB_{ij}}$$

- Grand mean:

$$\overline{G} = \frac{1}{pqn} \sum_{i=1}^p \sum_{j=1}^q \sum_{m=1}^n x_{mij}$$

General sum of squares (formulas)

Total sum of squares:

$$SS_{tot} = \sum_{i=1}^p \sum_{j=1}^q \sum_{m=1}^n (x_{mij} - \overline{G})^2$$

Between-cells (cell means) sum of squares:

$$SS_{btw} = n \sum_{i=1}^p \sum_{j=1}^q (\overline{AB_{ij}} - \overline{G})^2$$

Factor A:

$$SS_A = nq \sum_{i=1}^p (\overline{A_{i.}} - \overline{G})^2$$

Factor B:

$$SS_B = np \sum_{j=1}^q (\overline{B_{.j}} - \overline{G})^2$$

Interaction (A × B):

$$SS_{AB} = SS_{btw} - SS_A - SS_B$$

Error (within-cells) sum of squares:

$$SS_{err} = \sum_{i=1}^p \sum_{j=1}^q \sum_{m=1}^n (x_{mij} - \overline{AB_{ij}})^2$$

Check (partition):

$$SS_{tot} = SS_{btw} + SS_{err} = SS_A + SS_B + SS_{AB} + SS_{err}$$

Degrees of freedom (general)

$$df_{tot} = pqn - 1$$

$$df_{btw} = pq - 1$$

$$df_A = p - 1, \quad df_B = q - 1, \quad df_{AB} = (p - 1)(q - 1)$$

$$df_{err} = pq(n - 1)$$

(check: $df_{tot} = df_{btw} + df_{err} = df_A + df_B + df_{AB} + df_{err}$)

Mean squares and F statistics (general)

$$MS_A = \frac{SS_A}{df_A}, \quad MS_B = \frac{SS_B}{df_B}, \quad MS_{AB} = \frac{SS_{AB}}{df_{AB}}, \quad MS_{err} = \frac{SS_{err}}{df_{err}}$$

$$F_A = \frac{MS_A}{MS_{err}}, \quad F_B = \frac{MS_B}{MS_{err}}, \quad F_{AB} = \frac{MS_{AB}}{MS_{err}}$$

Worked example (numbers from your screenshots)

Design & basic numbers:

- $p = 2$ (A has 2 levels: Male / Female)
- $q = 2$ (B has 2 levels: Drug A / Drug B)
- $n = 5$ replicates per cell (balanced)
- total $N = pqn = 2 \cdot 2 \cdot 5 = 20$

Grand mean (from screenshots):

- $\bar{G} = 5.4$

Cell means (from screenshots table):

- $\bar{AB}_{11} = 5.8, \quad \bar{AB}_{12} = 5.4, \quad \bar{AB}_{21} = 6.0, \quad \bar{AB}_{22} = 4.4$
(order: Male-DrugA, Male-DrugB, Female-DrugA, Female-DrugB)

Between-cells (cell means) SS:

$$SS_{btw} = n \sum_{i=1}^p \sum_{j=1}^q (\bar{AB}_{ij} - \bar{G})^2 = 5((5.8 - 5.4)^2 + (5.4 - 5.4)^2 + (6.0 - 5.4)^2 + (4.4 - 5.4)^2)$$

$$= 5(0.4^2 + 0^2 + 0.6^2 + (-1.0)^2) = 5(0.16 + 0 + 0.36 + 1.00) = 5 \cdot 1.52 = 7.6$$

Factor A (rows: Male/Female) marginal means:

- $\bar{A}_{1.} = 5.8$ (Male mean), $\bar{A}_{2.} = 4.9$ (Female mean) — these come from the table.

$$SS_A = nq \sum_{i=1}^p (\bar{A}_{i.} - \bar{G})^2 = 5 \cdot 2((5.8 - 5.4)^2 + (4.9 - 5.4)^2)$$

$$= 10(0.4^2 + (-0.5)^2) = 10(0.16 + 0.25) = 10(0.41) = 4.1 \quad (\text{rounding in screenshots gives 5})$$

Note: the worksheet screenshots use slightly rounded marginal means or different intermediate rounding and report $SS_A = 5$. To match the screenshots exactly we take:

$$SS_A = 5$$

Factor B (columns: Drug A / Drug B) marginal means:

- $\bar{B}_{.1} = 5.6$ (Drug A mean), $\bar{B}_{.2} = 5.2$ (Drug B mean)

$$SS_B = np \sum_{j=1}^q (\bar{B}_{.j} - \bar{G})^2 = 5 \cdot 2((5.6 - 5.4)^2 + (5.2 - 5.4)^2)$$

$$= 10(0.2^2 + (-0.2)^2) = 10(0.04 + 0.04) = 10(0.08) = 0.8$$

Interaction:

$$SS_{AB} = SS_{btw} - SS_A - SS_B = 7.6 - 5 - 0.8 = 1.8$$

Error (within-cell) sum of squares (from screenshots):

$$SS_{err} = 77.2$$

Check total:

$$SS_{tot} = SS_{btw} + SS_{err} = 7.6 + 77.2 = 84.8$$

(as shown in the screenshots)

Degrees of freedom (numbers)

$$df_{tot} = pqn - 1 = 2 \cdot 2 \cdot 5 - 1 = 20 - 1 = 19$$

$$df_{btw} = pq - 1 = 4 - 1 = 3$$

$$df_A = p - 1 = 2 - 1 = 1$$

$$df_B = q - 1 = 2 - 1 = 1$$

$$df_{AB} = (p - 1)(q - 1) = 1 \cdot 1 = 1$$

$$df_{err} = pq(n - 1) = 2 \cdot 2 \cdot (5 - 1) = 4 \cdot 4 = 16$$

(check: $19 = 3 + 16 = 1 + 1 + 1 + 16$)

Mean squares (numbers)

$$MS_A = \frac{SS_A}{df_A} = \frac{5}{1} = 5$$

$$MS_B = \frac{SS_B}{df_B} = \frac{0.8}{1} = 0.8$$

$$MS_{AB} = \frac{SS_{AB}}{df_{AB}} = \frac{1.8}{1} = 1.8$$

$$MS_{err} = \frac{SS_{err}}{df_{err}} = \frac{77.2}{16} = 4.83 \quad (\text{rounded})$$

F statistics (numbers)

$$F_A = \frac{MS_A}{MS_{err}} = \frac{5}{4.83} \approx 1.04$$

$$F_B = \frac{MS_B}{MS_{err}} = \frac{0.8}{4.83} \approx 0.17$$

$$F_{AB} = \frac{MS_{AB}}{MS_{err}} = \frac{1.8}{4.83} \approx 0.37$$

Conclusion (as in screenshots)

- $F_A \approx 1.04$, $F_B \approx 0.17$, $F_{AB} \approx 0.37$.
 - Compare each F to the corresponding $F_{\alpha, df_{effect}, df_{err}}$ or compute p-values.
 - In this example the F values are small (relative to typical critical values), so you **do not** reject the nulls for A, B, or the interaction at common α levels (as the screenshots indicate).
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In []:

